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Test- 30/14

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Solution

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BPSC TEST-30/14 (SOLUTION)

Q 1. Ans. C

Explanation:

- **Pair 1 is correct: Article 94 - Vacation, resignation and removal of the Speaker**
  - ✓ Article 94 is designed to balance the security of tenure for the presiding officers with their accountability to the House. It ensures that the Speaker or Deputy Speaker cannot hold the house hostage, but also protects them from arbitrary removal.
  - ✓ **It explicitly lays out three conditions for vacating the office:**
    - If they cease to be a member of the Lok Sabha.
    - If they resign (the Speaker hands the resignation to the Deputy Speaker, and vice versa.
    - If they are removed by a resolution passed by a majority of all the then members of the Lok Sabha (this is known as an **Effective Majority**, not a simple or special majority). Such a resolution requires a 14-day prior notice.
- **Pair 2 is correct: Article 118 - Rules of procedure of a House of Parliament**
  - ✓ This article establishes the fundamental democratic principle of **Parliamentary Sovereignty in internal matters**. It ensures that the legislature is the absolute master of its own daily functioning, free from executive micromanagement.
  - ✓ **The Justification:** It allows each house to draft its own "Rules of Procedure and Conduct of Business." Crucially, Article 118(3) empowers the President (after consulting the Chairman of RS and Speaker of LS) to make rules for **Joint Sittings**. Under Article 118(4), it is constitutionally mandated that the Speaker of the Lok Sabha presides over these joint sittings.
- **Pair 3 is correct: Article 100** Quorum is a safeguard against "hijacking" by a minority. It ensures that legislative business and national decisions are not made by a tiny, unrepresentative handful of members who just happened to show up.
  - ✓ **The Justification:** The Constitution strictly fixes the quorum at **one-tenth (10%)** of the total membership of the House (including the presiding officer). That means a minimum of 55 members in the Lok Sabha and 25 in the Rajya Sabha must be present. If this number is not met, Article 100(4) places a mandatory duty on the presiding officer to either adjourn the House or suspend the meeting until there is a quorum.
- **Pair 4 is incorrect:**
  - ✓ **Article 119:** Article 119 is entirely different. It exists to prevent Parliament from indefinitely stalling the national budget. It allows Parliament to pass laws to strictly regulate financial procedures and timelines, ensuring that critical money matters don't get bogged down in endless debate.
  - ✓ **Article 122:** This enforces the **Separation of Powers**. It prevents the Judiciary from acting as a strict schoolmaster over the Legislature's internal day-to-day procedures. If a law is passed but a minor internal rule of the House was broken during the debate, the courts cannot invalidate the law on the grounds of "procedural irregularity."

Additional Facts:

- **Limits on Article 122 (Judicial Review):** While Article 122 protects Parliament from judicial interference over procedural irregularities, the Supreme Court (in landmark cases like Raja Ram Pal v. Hon'ble Speaker, Lok Sabha, 2007) has clearly ruled that parliamentary actions are **not immune** from judicial review if they suffer from substantive unconstitutionality, gross illegality, or denial of fundamental rights.
- **Article 93:** While Article 94 deals with the removal of the Speaker/Deputy Speaker, it is **Article 93** that mandates the Lok Sabha to choose two members to be Speaker and Deputy Speaker "as soon as may be" after a general election.
- **Casting Vote (Article 100(1)):** Under Article 100, the Speaker or Chairman does not vote in the first instance during a regular debate. They only exercise a **Casting Vote** in the case of a strict tie, ensuring that the legislative machinery does not grind to a halt due to a deadlock.
- **Pro Tem Speaker:** The Constitution does not explicitly mention the term "Pro Tem Speaker." The appointment of the Speaker Pro Tem (usually the senior-most member) to administer oaths to new MPs before a permanent Speaker is elected is a convention drawn from Article 99, conducted by the President.

Q 2. Ans. C

Explanation:

- **Committee on Government Assurances:** This committee's specific mandate is to track, scrutinise, and report on the extent to which promises, undertakings, and assurances given by Ministers on the floor of the House have been implemented in a timely and satisfactory manner.
- **Additional Facts:**
  - ✓ **Composition:** It operates separately in both Houses. The Lok Sabha committee consists of 15 members, while the Rajya Sabha committee consists of 10 members.
  - ✓ **Establishment:** The committee was first constituted in 1953 to ensure executive accountability to the legislature.
  - ✓ **Standard Timeframe:** As per parliamentary norms, any assurance given by a Minister on the floor of the House is expected to be fulfilled within three months.
  - ✓ **Committee on Public Undertakings (Distinction):** Often confused with options sounding like "Ministerial Undertakings," this actual committee consists of 22 members (15 from LS, 7 from RS) and is tasked exclusively with examining the reports, accounts, and autonomy of Public Sector Undertakings (PSUs).
  - ✓ **Committee on Subordinate Legislation (Distinction):** Another key accountability committee that examines whether the executive is properly exercising its delegated constitutional powers to frame rules, regulations, and bye-laws.

Q 3. Ans. B

Explanation:

- **Option B is not a Fundamental Duty)**The mandate to protect every monument or place or object of artistic or historic interest declared to be of national importance is actually a **Directive Principle of State Policy (DPSP)** under **Article 49**. It is an obligation placed on the State, not an individual citizen's Fundamental Duty.
- Under Article 51A(h), citizens are explicitly required to develop the scientific temper, humanism and the spirit of inquiry and reform.
- Under Article 51A(i), citizens are explicitly required to safeguard public property and to abjure violence.
- **Additional Facts:**
  - ✓ **Heritage vs. Monuments:** While protecting specific national monuments is a DPSP (Article 49), valuing and preserving the rich heritage of our composite culture is a Fundamental Duty (Article 51A(f)).
  - ✓ **Origin:** Fundamental Duties (Part IVA, Article 51A) were added by the **42nd Amendment Act (1976)** based on the recommendations of the **Swaran Singh Committee**. They were inspired by the Constitution of the USSR.
  - ✓ **11th Duty Addition:** Originally, there were 10 duties. The 11th duty, mandating parents/guardians to provide opportunities for education to children between 6 and 14 years, was added by the **86th Amendment Act (2002)**.
  - ✓ **Non-Justiciable:** Like DPSPs, Fundamental Duties are non-justiciable. The Constitution does not provide for their direct enforcement by the courts, although Parliament is free to enforce them via suitable legislation (e.g., The Prevention of Insults to National Honour Act, 1971).
  - ✓ **Applicability:** Unlike certain Fundamental Rights, which extend to all persons, Fundamental Duties apply **only to Indian citizens**, not foreigners.

Q 4. 4. Ans. A

Explanation:

- N. Vittal served as the **Chief Vigilance Commissioner (CVC)** of India from September 1998 to September 2002.
- **Additional Facts:**
  - ✓ **Vineet Narain Judgment:** He was the **first CVC** appointed after the landmark 1997 Vineet Narain Supreme Court ruling. This judgment mandated **statutory status** for the CVC to shield the institution from executive interference (later codified by the CVC Act, 2003).
  - ✓ He took the unprecedented and bold step of publishing the names of corrupt senior civil servants (IAS and IPS officers) facing departmental inquiries on the official CVC website.
  - ✓ Before his CVC tenure, as Secretary of the Department of Electronics and Telecommunications, he was instrumental in driving India's software and telecom revolution, notably setting up the **Software Technology Parks of India (STPI).**
  - ✓ He was awarded the **Padma Bhushan in 2012** for his distinguished civil service

Q 5. Ans. D

Explanation:

- **Rigidity of the Constitution is a federal feature:** A rigid constitution ensures that the fundamental structure and the division of powers between the Centre and the States cannot be unilaterally altered by the central government. In India, amending such crucial federal provisions requires not just a special majority in Parliament, but also ratification by at least half of the state legislatures (Article 368).
- **Additional Facts (Federal vs. Unitary Features):**
  - ✓ **Unitary Features (Centralising tendencies):** Single Constitution (states don't have their own), Integrated Judiciary (Supreme Court at the apex enforcing both state and central laws), Appointment of State Governors by the Centre, All-India Services (IAS, IPS), Emergency provisions, and a strong Centre.
  - ✓ **Federal Features (Decentralising tendencies):** Dual polity (Union and State governments), Written Constitution, Division of Powers (Seventh Schedule), Supremacy of the Constitution, Rigidity of the Constitution, Independent Judiciary, and Bicameralism.
  - ✓ **K.C. Wheare's Definition:** Because the Indian Constitution blends both, constitutional expert K.C. Wheare famously described India as "Quasi-Federal" (federal in form, unitary in spirit).

Q 6. Ans. B

Explanation:

- The Bharatiya Nyaya Sanhita (BNS), 2023, along with the other two new criminal law bills, formally received the assent of the President of India on **December 25, 2023**.
- **Additional Facts:**
  - ✓ **Enforcement Date:** While the BNS received presidential assent in December 2023, it officially came into effect across the country on **July 1, 2024**.
  - ✓ **Replaced Legislation:** The BNS, 2023, completely repealed and replaced the colonial-era **Indian Penal Code (IPC), 1860**.
  - ✓ **Companion Laws:** The other two laws that received assent on the same day are the **Bharatiya Nagarik Suraksha Sanhita (BNSS),** replacing the **CrPC, 1973,** and the **Bharatiya Sakshya Adhiniyam (BSA),** replacing the **Indian Evidence Act, 1872.**
  - ✓ The bill was passed by the Lok Sabha on December 20, 2023, and by the Rajya Sabha on December 21, 2023.

Q 7. Ans. B

Explanation:

- **Under the Delimitation Commission Act, 2002, the Commission is a three-member body consisting of:**
  - ✓ A serving or retired **Judge of the Supreme Court** as the Chairperson.
  - ✓ The **Chief Election Commissioner** or an Election Commissioner nominated by the CEC.
  - ✓ The **State Election Commissioner** of the concerned State(s) as an ex-officio member.
- **Additional Facts:**
  - ✓ **Judicial immunity:** The orders of the Delimitation Commission have the force of law and cannot be called into question before any court. This prevents election processes from being indefinitely stalled by litigation.
  - ✓ **No legislative override:** While its final orders are laid before the Lok Sabha and the respective State Legislative Assemblies for information, the legislatures are constitutionally barred from making any modifications to them.
  - ✓ **Associate members:** The Commission is assisted by Associate Members (up to 5 MPs and 5 MLAs from the concerned state) chosen by the respective Speakers. They provide local geographical and demographic insight but have absolutely **no voting rights** or power to sign the final decisions.
  - ✓ **Constitutional Basis:** Article 82 provides for the readjustment of constituencies after every census, while Article 170 deals with the delimitation of territorial constituencies for state legislative assemblies.
  - ✓ **Delimitation Commissions have been set up four times in the past:** 1952, 1963, 1973, and 2002. There was no delimitation after the 1981 and 1991 censuses.
  - ✓ Under the **84th Constitutional Amendment Act (2001),** the allocation of seats to states in the Lok Sabha was frozen based on the 1971 census until the relevant figures for the first census taken after the year 2026 have been published.
  - ✓ To initiate the unfreezing process, the Union Government introduced a package of bills in the Lok Sabha on **April 16, 2026**. The two most important were the **Delimitation Bill, 2026** and the **Constitution (131st Amendment) Bill, 2026**.

- ✓ **Key Proposed Changes:**
  - **Census Shift:** Instead of waiting for the results of the delayed upcoming census (Census 2027), the Delimitation Bill proposed using the **2011 Census** data to redraw electoral constituencies immediately.
  - **Seat Expansion:** The **131st Amendment Bill** proposed raising the maximum permissible strength of the Lok Sabha from 550 to 850 members.
  - The delimitation exercise is also practically linked to the implementation of the 33% Women's Reservation in the Lok Sabha and State Assemblies.
  - The Constitution (131st Amendment) Bill, 2026 **failed to pass** in the Lok Sabha. Because it was a constitutional amendment, it required a special majority (two-thirds of members present and voting). It received 298 votes, falling 54 votes short of the required 352.

Q 8. Ans. A

Explanation:

- **Statement 1 is correct:** The law grants a nominated member (e.g., in the Rajya Sabha) a six-month window to decide their political allegiance. If they join a political party within six months of taking their seat, they are not disqualified. However, if they join any political party after the expiry of this six-month period, they incur disqualification.
- **Statement 2 is incorrect:** The Speaker and Deputy Speaker of the Lok Sabha (and the Deputy Chairman of the Rajya Sabha, etc.) are explicitly granted an exemption under Paragraph 5 of the Tenth Schedule. To maintain the absolute impartiality and dignity of the presiding officer's chair, the law allows them to voluntarily give up the membership of their original political party upon election to the office. As long as they do not rejoin that party or join another party during their tenure, they are exempt from disqualification.
- **Statement 3 is correct:** Originally, the Tenth Schedule allowed an exemption if a "split" occurred, defined as one-third of the members of the legislature party defecting together. Because this loophole was heavily abused to engineer wholesale, legalised defections (often termed "Aaya Ram, Gaya Ram" politics), the **91st Constitutional Amendment Act, 2003,** completely deleted the "split" provision. Today, exemption is only granted in the case of a "merger," which requires at least two-thirds of the members to agree.
- **Statement 4 is incorrect:** In the landmark Kihoto Hollohan v. Zachillhu (1992) judgment, the Supreme Court upheld the constitutional validity of the Tenth Schedule but placed a strict limit on judicial intervention. The Court ruled that the Presiding Officer acts as a tribunal. Therefore, judicial review is strictly confined to the final decision of the Speaker/Chairman. Courts are constitutionally barred from intervening at the interlocutory stage (i.e., while the disqualification proceedings are still pending or before a final order is passed).
- **Additional Facts:**
  - ✓ The Anti-Defection Law (Tenth Schedule) was added to the Constitution by the **52nd Amendment Act, 1985**.
  - ✓ **An independent, unattached member (elected without a party ticket) faces strict rules:** they are disqualified if they join any political party after the election.
  - ✓ **Whip Violation:** A member can be disqualified not just for resigning from their party, but also for voting (or abstaining from voting) in the House contrary to the directions (the "Whip") issued by their political party without obtaining prior permission within 15 days.
  - ✓ **No Time Limit for Speaker:** The Tenth Schedule currently does not prescribe any specific time limit within which the Presiding Officer must decide on a disqualification petition, a loophole that has led to prolonged delays in various state assemblies. However, in Keisham Meghachandra Singh v. Hon'ble Speaker (2020), the Supreme Court advised that Speakers should ideally decide within a reasonable period of three months.

Q 9. Ans. B

Explanation:

- **Statement 1 is incorrect:** Under Section 25 of the Representation of the People Act, 1951, the statutory authority to provide a sufficient number of polling stations for a constituency lies directly with the **District Election Officer (DEO)**, not the Chief Electoral Officer (CEO), subject to the previous approval of the Election Commission of India.
- **Statement 2 is correct:** To ensure electoral accessibility, the ECI's manual on polling stations

explicitly mandates that they should be set up in a way that ordinarily no voter is required to travel more than **2 kilometres** to cast their vote.

- **Statement 3 is incorrect:** The upper ceiling on the maximum number of voters permitted per polling station is **not** codified in the Representation of the People Act, 1951. It is strictly an administrative guideline issued by the ECI, meaning the Commission can increase or decrease the cap flexibly without requiring a parliamentary amendment.
- **Statement 4 is correct:** In July 2025, during the Special Intensive Revision (SIR) of electoral rolls, **Bihar** became the first state in the country to successfully rationalise all its polling stations, permanently capping the maximum number of electors per booth at **1,200** to reduce voter congestion and long queues.
- **Additional Facts:**
  - ✓ To successfully implement the new 1,200-voter cap, Bihar had to create **12,817 entirely new polling stations**, raising the state's total booth count from **77,895 to 90,712.**
  - ✓ Historically (pre-2009), the standard ceiling was 1,200 electors. After the nationwide rollout of EVMs in 2009, the ECI raised the cap to 1,500. During the COVID-19 pandemic, the cap was temporarily slashed to 1,000 to ensure social distancing, before reverting to 1,500 for the 2024 General Elections. The 2025 move in Bihar marks a deliberate policy shift back to smaller booths.
  - ✓ **Link to 'One Nation, One Election' (ONOE):** Reducing the voter cap to 1,200 nationwide has massive logistical implications for the proposed simultaneous elections (ONOE). Smaller booth caps mean a significant surge in the total number of polling stations required across India, proportionately increasing the immediate demand and budgetary requirements for additional EVMs, VVPATs, and security personnel.
  - ✓ **100% Webcasting:** As a recent administrative milestone to ensure transparency, 100% of the polling stations during the recent Bihar assembly elections were equipped with live-webcasting facilities, allowing the central Election Commission to monitor booths in real-time.

Q 10. Ans. B

Explanation:

- **Statement 1 is incorrect (Article 243C):** The Constitution does not mandate direct election for the village Panchayat Chairperson. Instead, they are elected in a manner determined by the State Legislature. The members of the Panchayat, however, are directly elected.
- **Statement 2 is correct (Article 243H):** The State Legislature may, by law, authorise a Panchayat to levy, collect, and appropriate specific taxes, duties, tolls, and fees.
- **Statement 3 is incorrect (Article 243E):** A Panchayat reconstituted upon premature dissolution does not enjoy a full five-year term. It continues only for the remainder of the dissolved Panchayat's original term.
- **Statement 4 is correct (Article 243K):** The State Election Commissioner, appointed by the Governor, has security of tenure and can only be removed from office in the exact same manner and on the same grounds as a High Court Judge.
- **Additional Facts:**
  - ✓ **Article 243A (Gram Sabha):** Comprises all registered voters in the village electoral roll, performing functions determined by the state legislature.
  - ✓ **Article 243B (Three-Tier System):** Mandates Panchayats at village, intermediate, and district levels. States with a population under 20 lakhs are exempt from forming the intermediate level.
  - ✓ **Article 243D (Reservations):** Mandates reservation of seats for SCs and STs strictly proportional to their population. A minimum of 1/3rd of the total seats must be reserved for women (including SC/ST women).
  - ✓ **Article 243F (Disqualifications):** The minimum age to contest Panchayat elections is 21 years, distinct from the 25-year requirement for State Legislatures.
  - ✓ **Article 243I (State Finance Commission):** Constituted every five years by the Governor to review Panchayat finances.
  - ✓ **Article 243G (Eleventh Schedule):** Deals with the 29 functional items placed within the purview of Panchayats (e.g., agriculture, health, education).

Q 11. Ans. D

Explanation:

- Bathymetry is the study and mapping of the underwater depth and terrain of ocean, sea, or lake floors. It is essentially the underwater equivalent of topography.

- **Additional Facts:**
  - ✓ **Isobaths:** These are imaginary lines on a map connecting points of equal underwater depth (the marine equivalent of contour lines on land).
  - ✓ **Measurement Techniques:** Modern bathymetry primarily uses Sonar (Echo Sounding) and Satellite Altimetry (measuring micro-variations in the sea surface height caused by the gravitational pull of massive underwater features like ridges and trenches).
  - ✓ **Topography vs. Bathymetry:** While bathymetry measures depth below sea level, topography (or hypsometry) measures elevation above sea level.
  - ✓ **GEBCO:** The General Bathymetric Chart of the Oceans (GEBCO) is the definitive international project mapping the world's ocean floors, operating jointly under the IHO and the UN's Intergovernmental Oceanographic Commission (IOC).

Q 12. Ans. A

Explanation:

- **Statement 1 is correct:** The Western Ghats are a continuous and unbroken chain of mountains that can only be crossed through specific passes (such as the Thal Ghat, Bhor Ghat, and Pal Ghat). Conversely, the Eastern Ghats are highly discontinuous, irregular, and heavily dissected by the major rivers flowing into the Bay of Bengal.
- **Statement 2 is correct:** The Western Ghats are structurally much higher than the Eastern Ghats. The average elevation of the Western Ghats ranges between **900 and 1,600 meters**, whereas the Eastern Ghats are much lower, averaging an elevation of about **600 meters**.
- **Statement 3 is incorrect:** Most of the major east-flowing Peninsular rivers—including the **Godavari** (originates at Trimbakeshwar), **Krishna** (originates at Mahabaleshwar), and **Kaveri** (originates at Talakaveri)—actually originate from the water divide of the **Western Ghats**, not the Eastern Ghats. They flow eastward across the peninsula into the Bay of Bengal due to the natural eastward tilt of the Deccan Plateau.
- **Additional Facts:**
  - ✓ **Meeting Point:** The Western Ghats and the Eastern Ghats converge at the **Nilgiri Hills** along the border of Tamil Nadu, Kerala, and Karnataka.
  - ✓ **Highest Peaks:**
    - **Western Ghats: Anamudi** (2,695 meters) located in the Anaimalai Hills of Kerala, is the highest peak in the Western Ghats and the entirety of Peninsular India.
    - **Eastern Ghats: Jindhagada Peak/Arma Konda** (1,690 meters) in Andhra Pradesh is the highest peak of the Eastern Ghats.
  - ✓ **Rainfall Patterns:** The Western Ghats act as a barrier to the moisture-laden southwest monsoon winds, causing heavy **orographic rainfall** on their western slopes. The Eastern Ghats run almost parallel to the track of the southwest monsoon, receiving less rain during summer but substantial rainfall during the retreating northeast monsoon (winter months).
  - ✓ **Ecological Status:** The Western Ghats are recognised globally as one of the world's eight **biodiversity hotspots** and are a designated UNESCO World Heritage Site. Local names include Sahyadri in Maharashtra and Karnataka, Nilgiri in Tamil Nadu, and Annamalai and Cardamom hills in Kerala.

Q 13. Ans. C

Explanation:

- **The Nilgiri Biosphere Reserve is a major ecological hotspot spanning the tri-junction of three southern Indian states:** Tamil Nadu, Kerala, and Karnataka.
- It encompasses a massive, contiguous network of highly recognised protected areas. Specifically, it includes the **Wayanad, Nagarhole, Bandipur and Mudumalai**, as well as Mukurthi National Park and Silent Valley National Park.
- Because of its vast geographic spread across the Western Ghats and its rain-shadow regions, it features a striking ecological gradient ranging from thorny scrub and dry deciduous forests to moist deciduous, semi-evergreen, and lush tropical evergreen forests (like the Sholas).
- **Additional Facts:**
  - ✓ **India's First:** Established in 1986, the Nilgiri Biosphere Reserve was the very first biosphere reserve declared in India. In 2000, it also became the first in India to be included in the UNESCO World Network of Biosphere Reserves.
  - ✓ **Key Fauna:** It provides the largest contiguous habitat for the **Asian Elephant** and the **Bengal Tiger** in India. It is also the primary stronghold for the endangered **Lion-tailed Macaque** and the **Nilgiri Tahr** (the state animal of Tamil Nadu).

- ✓ **Indigenous Communities:** The reserve is home to several distinct tribal groups with rich cultural heritage, most notably the **Todas** (famous for their barrel-shaped huts and unique embroidery), Kotas, Kurumbas, and Irulas.
- ✓ **Other Options Evaluated:**
  - **Agasthyamalai Biosphere Reserve:** Located further south, spanning only Kerala and Tamil Nadu. It includes the Neyyar, Peppara, and Shendurney Wildlife Sanctuaries.
  - **Seshachalam Biosphere Reserve:** Located entirely in Andhra Pradesh (Eastern Ghats) and is globally famous for its endemic population of Red Sandalwood (Pterocarpus santalinus).

Q 14. Ans. C

Explanation:

- The world's highest tidal range is observed in the **Bay of Fundy**, located on the Atlantic coast of North America, between the Canadian provinces of New Brunswick and Nova Scotia.
  - ✓ The tidal range in the Bay of Fundy can reach 16 meters (over 50 feet). This occurs due to a combination of the bay's unique funnel shape, its depth, and a phenomenon called "tidal resonance" (the time it takes for a large wave to travel from the mouth of the bay to the opposite end matches the time of the ocean tide).
- **Gulf of Kutch:** While the Gulf of Kutch does not hold the world record, it is notable for having the highest tidal range in **India**, reaching up to 8 meters.
- **Bay of Bengal & Gulf of Mannar:** The Bay of Bengal experiences significant tides (especially near the Sundarbans and Hooghly estuary), but they do not match the extremes of the Bay of Fundy. The Gulf of Mannar generally experiences much lower tidal ranges.

Q 15. Ans. B

Explanation:

- **Statement 1 is correct:** El Niño is characterised by a significant weakening (and sometimes a reversal) of the trade winds, which inherently weakens the Walker Circulation. Because the warm water pool shifts eastward, the western equatorial Pacific (near Indonesia and Australia) experiences relatively cooler waters and anomalous **high atmospheric pressure**, leading to dry, drought-like conditions.
- **Statement 2 is incorrect:** During La Niña, the trade winds become unusually strong, pushing more warm surface water westward toward Asia. As a result, the oceanic thermocline becomes shallower (it rises) off the coast of Peru in the eastern Pacific. This **enhances** the upwelling of cold, nutrient-rich water from the deep ocean. The deepening of the thermocline and the suppression of upwelling actually occur during an El Niño event.
- **Statement 3 is correct:** Since tropical convection (rising warm air and thunderstorm formation) follows the warmest ocean waters, the eastward shift of the warm water pool during El Niño drags the primary zone of heavy precipitation along with it. This moves rainfall away from Indonesia and the western Pacific toward the central and eastern Pacific.
- **Additional Facts:**
  - ✓ **Southern Oscillation Index (SOI):** This is the atmospheric component of ENSO. It measures the difference in surface air pressure between **Tahiti** (central/eastern Pacific) and **Darwin, Australia** (western Pacific). During an El Niño event, the SOI is **negative** (pressure is unusually high in Darwin and low in Tahiti).
  - ✓ **Impact on Indian Monsoon:** Historically, strong El Niño years are strongly correlated with suppressed or deficient Southwest Monsoon rainfall in India. Conversely, La Niña years generally favour a robust, above-average monsoon.
  - ✓ **El Niño Modoki:** A distinct variation of the phenomenon where maximum ocean warming occurs in the **central equatorial Pacific** rather than the eastern Pacific. It is flanked by anomalously cooler waters to both the east and west, which alters global teleconnections differently than a traditional El Niño.
  - ✓ **Peruvian Anchovy Fishery:** The upwelling off the coast of Peru supports one of the world's largest fisheries. During El Niño, the suppression of this cold, nutrient-rich upwelling causes massive die-offs of phytoplankton and a subsequent collapse of the local anchovy catch.

Q 16. Ans. C

Explanation:

- **Zagros Mountains — Iran (Correct):** The Zagros Mountains form the largest mountain range in Iran, stretching across the western and southwestern parts of the country into northeastern Iraq. They are fold mountains formed by the collision of the Arabian and Eurasian plates.

- **Drakensberg — South Africa (Correct):** The Drakensberg is the highest and most extensive mountain range in Southern Africa, located primarily in South Africa and enclosing the landlocked kingdom of Lesotho.
- **Apennine Mountains — Spain (Incorrect):** The Apennine Mountains run along the entire length of peninsular **Italy**, forming the structural backbone of the country. Spain's prominent mountain ranges include the Pyrenees (bordering France) and the Sierra Nevada.
- **Arakan Yoma — Myanmar (Correct):** Also known as the Rakhine Mountains, this range lies in western Myanmar, acting as a natural geographic barrier between the coast of Rakhine State and the central Irrawaddy River basin.
- **Additional Facts:**
  - ✓ **Highest Peaks of these Ranges:**
    - **Zagros Mountains:** Mount Dena (~4,409 m)
    - **Drakensberg:** Thabana Ntlenyana (~3,482 m, located in Lesotho)
    - **Apennines:** Corno Grande (~2,912 m)
    - **Arakan Yoma:** Mount Victoria (Nat Ma Taung, ~3,053 m)
  - ✓ **Atlas Mountains:** Located in Northwest Africa, spanning across **Morocco, Algeria, and Tunisia**, separating the Mediterranean and Atlantic coastlines from the Sahara Desert.
  - ✓ **Ural Mountains:** A major north-south mountain range running through western **Russia** that forms the traditional, natural boundary between Europe and Asia.
  - ✓ **Southern Alps:** A majestic mountain range that runs along the entire length of the South Island of **New Zealand** (highest peak is Aoraki/ Mount Cook). Whereas the European Alps span France, Switzerland, Italy, Austria, etc.

Q 17. Ans. D

Explanation:

- The Katanga province, located in the southern Democratic Republic of the Congo (DRC), is world-renowned for its vast, high-grade deposits of **copper** and **cobalt**.
- Geographically, it forms the heart of the **Central African Copperbelt**, a massive geological structure that stretches from northern Zambia into the DRC.
- Because cobalt is primarily extracted as a by-product of copper mining in this region, Katanga's copper mines are simultaneously responsible for making the DRC the world's undisputed leader in cobalt production, supplying over two-thirds of the global market demand.
- **Additional Facts:**
  - ✓ **Bauxite (Aluminium Ore):** Australia is the top global producer, while Guinea (West Africa) holds the world's largest proven reserves.
  - ✓ **Iron Ore:** The **Pilbara region** in Western Australia and the **Carajás Mine** in Brazil are among the largest and most significant iron ore mining centres in the world.
  - ✓ **Coal:** The **Ruhr Valley** (Germany) and the **Appalachian region** (USA) are historically famous coal regions. Globally, China and India are the dominant producers and consumers of coal today.
  - ✓ **Copper (Outside Africa): Chile** remains the world's largest producer of copper, largely driven by massive open-pit mines like **Chuquicamata** and **Escondida** located in the Atacama Desert.
  - ✓ **Lithium (The Green Mineral):** South America's **Lithium Triangle**, comprising high-altitude salt flats in **Chile, Argentina, and Bolivia**, holds more than half of the world's lithium resources, heavily sought after for electric vehicle (EV) batteries.

Q 18. Ans. C

Explanation:

- The **Bab-el-Mandeb Strait** is historically known as the "Gate of Tears" or "Gate of Grief" in Arabic.
- This name stems from the inherent dangers of navigating its currents, unpredictable winds, and shallow reefs. According to Arab legend, the name also refers to the immense loss of life during a massive ancient earthquake that separated the Arabian Peninsula from the Horn of Africa.
- Geographically, it acts as a critical strategic chokepoint connecting the Red Sea to the Gulf of Aden (and the wider Indian Ocean), situated between Yemen on the Arabian Peninsula and Djibouti and Eritrea in the Horn of Africa.
- **Additional Facts:**
  - ✓ **Strait of Hormuz:** Located between Iran and Oman, it connects the Persian Gulf to the Gulf of Oman and the Arabian Sea. It is arguably the world's most critical chokepoint for the global transit of crude oil.
  - ✓ **Strait of Malacca:** The primary shipping lane between the Indian and Pacific Oceans, passing between the Malay Peninsula (Malaysia/ Singapore) and the Indonesian island of Sumatra.
  - ✓ **Strait of Magellan:** A navigable sea route in

southern Chile separating mainland South America from Tierra del Fuego, historically providing a passage between the Atlantic and Pacific Oceans before the Panama Canal was built.

- ✓ **Strait of Gibraltar:** Connects the Atlantic Ocean to the Mediterranean Sea and separates the Iberian Peninsula (Spain/Gibraltar) from North Africa (Morocco).
- ✓ **Bosphorus Strait:** Located entirely within Turkey, it forms part of the continental boundary between Europe and Asia, and connects the Black Sea to the Sea of Marmara.

Q 19. Ans. C

Explanation:

- **Pair 1 is correct (Debrigarh):** Located in **Odisha** along the Hirakud Reservoir.
  - ✓ Odisha formally proposed upgrading it to "Debrigarh-Gaur National Park" due to its exceptionally high density of Indian gaurs (bison).
- **Pair 2 is incorrect (Kukrail):** Located in Lucknow, **Uttar Pradesh**.
  - ✓ The Supreme Court recently granted the UP government conditional clearance to build India's first urban "Night Safari and Zoological Park" within this reserve.
- **Pair 3 is correct (Guru Ghasidas-Tamor Pingla):** Located in **Chhattisgarh**.
  - ✓ Recently notified as a unified Tiger Reserve (Chhattisgarh's fourth), it was highlighted in the NTCA's 2026 STRIDES report as a key focus area for relocating villages from core tiger habitats.
- **Pair 4 is incorrect (Madhav National Park):** Located in Shivpuri, **Madhya Pradesh**.
  - ✓ It was officially elevated to become India's 58th Tiger Reserve. It also houses the Sakhya Sagar Ramsar site.
- **Additional Facts:**
  - ✓ **Patna Bird Sanctuary (Uttar Pradesh):** Newly designated as a Ramsar Wetland, contributing to India's updated tally of 98 Ramsar sites in 2026.
  - ✓ **Chhari-Dhand (Gujarat):** Also recently added to India's growing list of recognised Ramsar Wetlands.
  - ✓ **Indravati & Udanti-Sitanadi (Chhattisgarh):** Along with Veerangana Durgavati (MP), these reserves were prominently featured in the 2026 NTCA report for having the highest number of villages still awaiting relocation from critical tiger habitats.

Q 20. Ans. D

Explanation:

- **Statement A is correct:** Jet streams are fast-flowing, narrow bands of wind. They are geostrophic winds because they flow parallel to the isobars, resulting from a balance between the pressure gradient force and the Coriolis force, and are located in the upper troposphere (around 9 to 14 km altitude).
- **Statement B is correct:** Driven by the Earth's rotation (the Coriolis effect) and global temperature differences, the primary jet streams flow predominantly from west to east in both the Northern and Southern Hemispheres.
- **Statement C is correct:** Jet streams rarely flow in a straight line. As they interact with the Earth's topography and differing air masses, they develop large, meandering loops and curves known as Rossby waves, which dictate large-scale weather systems.
- **Statement D is incorrect:** The wind speed of a jet stream is directly determined by the temperature contrast between the interacting air masses (such as cold polar air and warm tropical air). Because this temperature gradient is steepest and most extreme during the winter months, jet stream velocities actually reach their maximum during the winter, not the summer.
- **Additional Facts:**
  - ✓ **Primary Types:** The two main jet streams in each hemisphere are the **Polar Jet Stream** (located around 50°–60° latitude) and the **Subtropical Jet Stream** (located around 30° latitude). The Polar jet is generally the stronger of the two.
  - ✓ **The Tropical Easterly Jet:** While jet streams generally flow west to east, the Tropical Easterly Jet is a unique, seasonal exception. It develops during the summer months in the upper troposphere over the Tibetan Plateau and flows east to west across India and Africa. It plays a highly crucial role in driving the sudden burst of the **Indian Southwest Monsoon**.
  - ✓ **Western Disturbances:** The Subtropical Westerly Jet stream is responsible for bringing "Western Disturbances", extratropical storm systems originating in the Mediterranean Sea, into the Indian subcontinent during winter. These disturbances cause crucial winter rainfall in northwestern India, which is vital for Rabi crops like wheat.
  - ✓ **Aviation Impact:** Airlines often utilise the west-to-east flow of jet streams to their advantage. Flying eastward with the jet stream significantly reduces flight time and fuel consumption, while flying westward against it increases flight time.

Q 21. Ans. B

Explanation:

Purnia does not share an international boundary with Nepal, whereas **Madhubani, Araria and Sitamarhi** do. Bihar shares an international border of about **729 km with Nepal**, extending across **seven districts—West Champaran, East Champaran, Sitamarhi, Madhubani, Supaul, Araria and Kishanganj**. Purnia lies immediately south of Araria and Kishanganj and therefore remains an inland district without an international boundary. It is bordered by **Araria, Katihar and Madhepura** within Bihar. The Indo-Nepal border in Bihar is an open international border governed under the **India-Nepal Treaty of Peace and Friendship (1950)**, allowing free movement of citizens of both countries. Because of this border, the northern districts play an important role in trade, cultural exchange, disaster management and border security.

Q 22. Ans. A

Explanation:

Dr. Rajendra Prasad was one of the foremost leaders associated with the Champaran Satyagraha of 1917, which marked Mahatma Gandhi's first successful experiment with **Satyagraha in India**. When Gandhi arrived in Champaran at the request of **Raj Kumar Shukla**, Rajendra Prasad, along with Brajkishore Prasad, Mazharul Haque, J.B. Kripalani and other local leaders, assisted him in conducting detailed investigations into the grievances of indigo cultivators. These peasants were suffering under the exploitative **Tinkathia System**, which compelled them to cultivate indigo on three kathas out of every twenty kathas of land. Rajendra Prasad coordinated volunteers, organised legal assistance, collected testimonies from thousands of peasants and helped publicise the movement. The success of the inquiry led to the **Champaran Agrarian Act of 1918**, which abolished many oppressive practices of European indigo planters. The movement transformed Gandhi into a national leader and also marked Rajendra Prasad's emergence as one of the foremost leaders of the Indian freedom struggle.

Q 23. Ans. C

Explanation:

- **Assertion (A) is true: Zone I (North-West Alluvial Plain)** receives relatively **lower rainfall** than **Zone II (North-East Alluvial Plain)** and is mainly drained by the **Gandak, Burhi Gandak, and Ghaghara** river systems. Although floods occur frequently, inundation generally recedes faster in many areas. In contrast, **Zone II** receives the **highest rainfall in Bihar** and is traversed by highly sediment-laden rivers such as the **Kosi, Bagmati, Kamla-Balan, Adhwara group, and Mahananda**. These rivers frequently overflow and create **extensive and prolonged waterlogging**, especially in low-lying areas. Hence, Zone II is generally more prone to prolonged waterlogging than Zone I.
- **Reason (R) is False:** Rivers in **Zone II**, particularly the **Kosi**, **Bagmati**, **Kamla-Balan**, and the **Adhwara group**, are among the **most unstable rivers** in Bihar. They carry heavy sediment loads, frequently change channels, breach embankments, and cause severe flooding and waterlogging. Their **high channel instability** is one of the major reasons for recurrent floods in north-eastern Bihar.

Q 24. Ans. B

Explanation:

The **Kisan Sabha organised at Munger in 1922** was led by **Sri Krishna Singh and Mohammad Zubair**, making option (B) the correct answer. This organisation represented one of the earliest organised efforts to mobilise peasants in Bihar against excessive rent, illegal cesses and exploitative zamindari practices. It laid the foundation for the later peasant movement in the province. **Swami Sahajanand Saraswati**, although regarded as the greatest peasant leader of Bihar, became associated with organised peasant politics later and founded the **Bihar Provincial Kisan Sabha in 1929**. This organisation subsequently evolved into the **All India Kisan Sabha (1936)**, which became one of the most influential peasant organisations in colonial India.

Q 25. Ans. A

Explanation:

Kunwar Singh fought his final battle against Captain Le Grand at Jagdishpur in April 1858. After successfully crossing the Ganga despite being seriously wounded, Kunwar Singh returned to Jagdishpur and recaptured his ancestral estate from British control. Captain Le Grand marched with British troops to suppress the rebellion but suffered a decisive defeat at the hands of Kunwar Singh's forces. Although already around **80 years of age**, Kunwar Singh displayed exceptional military leadership during the engagement. According to historical accounts preserved in official

records, he had sustained a severe bullet injury while crossing the Ganga and is believed to have severed his injured wrist before offering it to the river as an act of sacrifice. He died shortly after his victory on **26 April 1858**, making this battle his last and one of the most celebrated victories of the Revolt of 1857 in eastern India. British officers **Douglas, Eyre and Lugard** fought Kunwar Singh during earlier phases of the rebellion, but **Captain Le Grand** commanded the British force in his final battle.

Q 26. Ans. D

Explanation:

- **Assertion (A) is False:** The agro-climatic zones of Bihar are **not created on the basis of administrative convenience**. Although, for ease of implementation, **entire districts are grouped under a particular zone**, the zoning itself is based on **natural resource characteristics** rather than district boundaries. The Government of Bihar classifies the state into **four agro-climatic zones (I, II, IIIA and IIIB)** according to differences in agro-ecological conditions.
- **Reason (R) is true:** Agro-climatic zoning is determined primarily by climatic and edaphic characteristics rather than administrative convenience.
  - ✓ **Government of Bihar classifies agro-climatic zones using factors such as:**
  - ✓ **Rainfall distribution**
  - ✓ **Temperature regime**
  - ✓ **Soil (edaphic) characteristics**
  - ✓ **Topography and physiography**
  - ✓ **Hydrology and drainage**
  - ✓ **Length of growing period**
- These natural factors determine crop suitability and agricultural planning, which is the fundamental purpose of agro-climatic zoning.

Q 27. Ans. C

Explanation:

Saltpetre (Potassium Nitrate, KNO_3) was one of the most valuable export commodities of early colonial India. During the seventeenth and eighteenth centuries, **Bihar was the principal producer of saltpetre in the world.** Districts around **Patna, Tirhut, Saran, Gaya and Shahabad** became famous for its manufacture. The English East India Company established a monopoly over the saltpetre trade because it was essential for manufacturing **gunpowder**. Patna emerged as one of the largest export centres.

Q 28. Ans. C

Explanation:

According to the **Bihar Economic Survey 2025–26, Agriculture, Forestry and Fishing** together contributed **23.1% of Bihar's Gross State Value Added (GSVA)** in **2023–24 (Provisional Estimates)**, reflecting the continued importance of the primary sector in the state's economy. Within this sector, **crop production contributed 9.5%** of GSVA, followed by **livestock at 8.1%**, while the remaining share came from forestry and fishing activities. Hence, **23.1%** is the correct answer.

Q 29. Ans. D

Explanation:

Dr. Rajendra Prasad served as the President of the Bihar Provincial Congress Committee during 1928–29, when the Indian National Congress was intensifying its campaign against British colonial rule. Under his leadership, the Congress organisation in Bihar was significantly strengthened through village-level committees, constructive programmes and increased public participation. This period coincided with the growing demand for **Poorna Swaraj**, the opposition to the **Simon Commission (1928)** and preparations for the **Civil Disobedience Movement** launched in 1930. Rajendra Prasad worked closely with national leaders such as Mahatma Gandhi, Sardar Vallabhbhai Patel and Jawaharlal Nehru while simultaneously nurturing regional leaders like Sri Krishna Singh and Anugrah Narayan Sinha. His organisational abilities made Bihar one of the most active provinces in the national movement, and he later became the **first President of independent India's Constituent Assembly** and the **first President of the Republic of India (1950–1962).**

Q 30. Ans. C

Explanation:

- **Statement 1 is correct:** The **Bagmati River originates in the Shivpuri Range of hills in Nepal**, approximately **16 km northeast of Kathmandu**, at an elevation of about **1,500 metres above mean sea level**. Initially, the river flows westward, draining the **Kathmandu Valley**, before cutting through the **Mahabharata Range** and turning south towards the Indo-Nepal border. The upper catchment of the Bagmati is of immense cultural and hydrological significance, as the river flows past the **Pashupatinath Temple**, one of the holiest

Hindu shrines. The Bagmati is also one of the major transboundary rivers shared by Nepal and India.

- **Statement 2 is incorrect: Bagmati flows for a much longer distance in Bihar than in Nepal.** The river has a **total length of about 589 km**, of which approximately **195 km lies in Nepal** and **394 km flows through Bihar**. Thus, nearly **two-thirds of the river's length lies within Bihar**. After entering India at **Shorwatia village in Sitamarhi district**, the river traverses the districts of **Sitamarhi, Muzaffarpur, Darbhanga, Samastipur and Khagaria** before ultimately joining the **Kosi River near Badlaghat**. Consequently, the river's Indian reach is substantially longer than its Nepali course.
- **Statement 3 is correct:** After traversing the Mahabharata hills, the **Bagmati enters Bihar in Sitamarhi district near the village of Shorwatia**. Once inside Bihar, the river exhibits a highly unstable and meandering course owing to the extremely gentle slope of the North Bihar plains. Frequent channel shifting, flooding and avulsion are characteristic features of the Bagmati basin. During its journey through Bihar, the river receives several important tributaries, including the **Lalbakeya**, **Darbhanga-Bagmati**, **Hasanpur-Bagmati** and the **Old Kamla**. After the confluence with the **Darbhanga-Bagmati near Hayaghat**, the river is locally known as the **Kareh**, before finally joining the **Kosi River**.

Q 31. Ans. B

Explanation:

- **Statement 1 is incorrect:** The river originates near the **Nepal–Tibet border, northeast of the Dhaulagiri massif**, at an elevation of about **7,620 metres** in the High Himalayas. The river is glacier-fed and forms part of the Himalayan river system. During its upper course, it flows through deep mountain gorges before descending into the plains. Therefore, attributing its origin to the Mahabharata Range is incorrect. The Mahabharata Range is a middle mountain range of Nepal, whereas the Gandak originates much farther north in the Greater Himalaya.
- **Statement 2 is correct:** In Nepal, the Gandak is popularly known as the **Kali** or **Kali Gandaki**. It is one of Nepal's major Himalayan rivers and flows through the famous **Kali Gandaki Gorge**, regarded as one of the deepest river gorges in the world. During its course in Nepal, the river receives several important tributaries, including the **Mayangadi**, **Bari**, and **Trisuli**, before reaching the Indo–Nepal border. Just above **Valmikinagar**, at **Triveni**, it is joined by the **Panchnad** and **Sohna** rivers.
- **Statement 3 is correct:** After travelling nearly **100 km through Nepal**, the Gandak enters the plains of Bihar near **Valmikinagar** in **West Champaran district**. It then flows generally southward, forming the **Uttar Pradesh–Bihar boundary** for a short distance before passing through the districts of **West Champaran, Saran, Muzaffarpur and Vaishali**. Finally, it joins the **left bank of the Ganga opposite Patna city**. The river has a total length of about **630 km**, of which approximately **370 km lies in Nepal and Tibet**, while the remaining course is in India.

Q 32. Ans. C

Explanation:

Bihar accounts for approximately **17.2% of India's total flood-prone area**, making it one of the most flood-affected states in the country. Nearly **76% of the population of North Bihar** lives under the recurring threat of floods. The state's flood problem is mainly caused by **Himalayan rivers**, such as the **Kosi, Gandak, Burhi Gandak, Bagmati, Kamla Balan, Mahananda and the Adhwara group of rivers**, which carry enormous water discharge and heavy sediment loads into the plains of Bihar. Around **65% of the catchment area of these rivers lies in Nepal and Tibet**, giving Bihar's flood management an international dimension. In addition to river floods, the state also faces severe **waterlogging** due to silted rivers, encroached drainage channels, embankments and natural depressions known as **Chaurs**. The **Kosi River**, often called the **"Sorrow of Bihar,"** is particularly notorious for frequent channel shifts and devastating floods.

Q 33. Ans. B

Explanation:

In **medieval Bihar**, **qasbas** emerged as **small urban centres** that served as **market towns, administrative headquarters, and centres of crafts and trade**. They connected rural production with regional markets and often housed government officials, merchants, artisans, and religious scholars. Their growth reflected the expansion of commerce, urbanisation, and administrative control under the Delhi Sultanate and the Mughals.

Q 34. Ans. B

Explanation:

The **Amil (or Amalguzar)** was the **principal revenue officer** at the **pargana** level during the Mughal period.

He was responsible for **assessing land revenue, collecting revenue from cultivators and zamindars, supervising revenue administration, and ensuring that the state's revenue demand was realized**. He worked with officials such as the **Qanungo** (custodian of land records) and the **Patwari** (village record keeper).

Q 35. Ans. A

Explanation:

- **Assertion (A) is true:** The **Fourth Agriculture Road Map (2023–2028)** of the Government of Bihar places much greater emphasis on **climate-resilient agriculture** than the earlier Road Maps. It focuses on climate-smart farming, promotion of stress-tolerant crop varieties, efficient water management, micro-irrigation, crop diversification, soil health management, integrated farming systems, and weather-based advisory services to make agriculture more resilient and sustainable.
- **Reason (R) is true:** Bihar has experienced an increasing frequency of **climate extremes**, including recurrent floods in north Bihar, droughts in several south Bihar districts, erratic monsoon rainfall, heat waves, cold waves, and unseasonal rainfall, all of which adversely affect agricultural productivity and farmers' livelihoods. These emerging climatic challenges are the primary reason why the Fourth Agriculture Road Map gives greater priority to climate-resilient agriculture. Therefore, the **Reason correctly explains the Assertion**.

Q 36. Ans. D

Explanation:

- The **Ugna Mahadev Mandir** is a revered **Lord Shiva temple** situated in **Bhawanipur village** of **Madhubani district, Bihar**, about **14 km from Madhubani town** and **180 km from Patna**. The temple holds immense religious and cultural significance in the **Maithili tradition**, as it is believed to be the sacred site where **Lord Shiva revealed his divine identity to the celebrated 14th-century Maithili poet Vidyapati**. According to local tradition, Lord Shiva served Vidyapati in the guise of a human attendant and once miraculously quenched his thirst by producing **Ganga water (Ganga-jal)**. When Shiva's true identity was inadvertently revealed during a household incident, he disappeared and later manifested as a **Shivling** at the same location, giving rise to the present-day shrine.
- To promote religious tourism, the **Government of Bihar** has proposed a **Shiva Temple Tourism Circuit**, aimed at improving infrastructure and pilgrim facilities at major Shaivite pilgrimage centres across the state. The proposed circuit includes **Ugna Mahadev Mandir (Madhubani)** along with other prominent shrines such as **Ajgaibinath Dham (Bhagalpur), Budhanath Temple, and Harihar Nath Temple**.

Q 37. Ans. B

Explanation:

- The **7th edition of the Khelo India Youth Games (KIYG) 2025**, the flagship youth sporting event under the Government of India's **Khelo India** initiative, was held from **4 to 15 May 2025**, with **Bihar** hosting the Games for the first time. The official **theme** of the event was **"Khel Ke Rang, Bihar Ke Sang."**
- The Games were conducted across **five venues in Bihar—Patna, Rajgir, Begusarai, Gaya, and Bhagalpur**—while the events of **gymnastics, shooting, and cycling** were organized in **New Delhi**. **Gajasimha** served as the **official mascot** of the Games. The 12-day sporting extravaganza featured competitions in **27 sports**, with athletes competing for a total of **285 gold medals**.
- As the host state, **Bihar** delivered its best-ever performance in the history of the Khelo India Youth Games by winning **36 medals**, including **7 gold, 11 silver, and 18 bronze medals**. This outstanding performance elevated Bihar from **21st position in the sixth edition** to **15th place** in the medal standings, marking the state's **highest-ever medal tally** in the competition.

Q 38. Ans. C

Explanation:

The **first Bihar Economic Survey** was published for the **financial year 2006–07** and was **tabled in the Bihar Legislative Assembly on 6 March 2007**. It was presented by **Sushil Kumar Modi**, who was serving as the **Deputy Chief Minister and Finance Minister of Bihar**. This was the **first occasion in the history of the Bihar Legislature** that a comprehensive assessment of the state's economic performance was formally placed before the Assembly. The survey was introduced to provide an evidence-based analysis of Bihar's economy, promote informed public discussion on development and governance, and support policy formulation. Since then, the **Bihar Economic Survey has become an annual pre-budget document**, presented before the state budget to review the state's economic progress and key socio-economic indicators.

Q 39. Ans. A

Explanation:

The **Bihar State Milk Co-operative Federation (COMFED)** achieved a major milestone in **2024–25** by exporting its **first international consignment of dairy products to the United States (USA) and Canada**. This marked COMFED's entry into the global dairy market and reflected the growing recognition of Bihar's dairy products abroad. The export was made under COMFED's well-known **'Sudha'** brand, which has expanded its presence from domestic markets to international destinations. Therefore, the **USA and Canada** are the correct answer

Q 40. Ans. C

Explanation:

According to the **Bihar Economic Survey 2025–26**, the **Credit-Deposit (CD) ratio of Scheduled Commercial Banks in Bihar stood at 53.5% as on 31 March 2025**, reflecting a steady improvement in credit flow within the state. During 2024–25, **bank deposits increased by 10.9% to ₹7.77 lakh crore**, while **bank credit grew by 17.6% to ₹4.16 lakh crore**. As a result, the CD ratio improved from **50.4% on 31 March 2024** to **53.5% on 31 March 2025**, indicating that banks are lending a larger proportion of the deposits mobilised in Bihar. However, the Survey also notes that the CD ratio remains below the national average, highlighting the need for further expansion of institutional credit to support economic growth. Therefore, **53.5%** is the correct answer.

Q 41. Ans. A

Explanation:

- **Statement 1 is correct:** According to the **Bihar Economic Survey 2025–26**, citing the **India State of Forest Report (ISFR) 2023**, Bihar's **total forest and tree cover** was **9,902.66 sq. km (approximately 9,903 sq. km)** in 2023. This includes **7,530.66 sq. km of forest cover** and **2,372 sq. km of tree cover**, confirming that the combined forest and tree cover was about **9,903 sq. km**.
- **Statement 2 is incorrect:** The Bihar Economic Survey shows that the **Very Dense Forest (VDF)** area **increased**, rather than declined, between **2011 and 2023**. The VDF area rose from **248 sq. km in ISFR 2011** to **333.72 sq. km in ISFR 2023**, indicating an improvement in the state's dense forest cover over the period. Therefore, the statement that the VDF area declined is incorrect.

Q 42. Ans. C

Explanation:

- **Statement 1 is incorrect:** Dr. **Anugrah Narayan Sinha** did **not** become the **second** Deputy Chief Minister of Bihar. He became the **first Deputy Chief Minister and Finance Minister of Bihar** in **1946** and continued in these positions until his death in **1957**. He played a pivotal role in laying the foundations of modern Bihar and is popularly remembered as **"Bihar Vibhuti."**
- **Statement 2 is correct:** Dr. Anugrah Narayan Sinha was a **member of the Constituent Assembly of India**, which was entrusted with framing the **Constitution of India**. After the Constitution came into force on **26 January 1950**, he also served in the **first Parliament of independent India**. His contribution to nation-building extended beyond Bihar to the constitutional development of the country.
- **Statement 3 is incorrect:** Dr. Anugrah Narayan Sinha **never served as the President of the Indian National Congress**. In **1923**, following his successful organisation of the **Gaya Congress Session (1922)**, he was appointed as **one of the General Secretaries of the All India Congress Committee (AICC)**. Thus, while he held an important organisational position within the Congress, he was **not the party's President**.

Q 43. Ans. C

Explanation:

- **Statement 1 is correct:** Bihar is the **principal holder of India's pyrite resources**, accounting for **about 94% of the country's total pyrite resources**. The major pyrite deposits are located in the **Amjhore region of Rohtas district**. Although commercial mining has been limited, Bihar continues to dominate India's pyrite resource base.
- **Statement 2 is correct:** significant **limestone deposits** occur in the districts of **Kaimur (Bhabua), Munger, and Rohtas**. Limestone is an important raw material for the **cement, lime, and construction industries**.

Q 44. Ans. D

Explanation:

- All four books listed are **novels written by Phanishwar Nath Renu (1921–1977)**, one of the foremost figures of modern Hindi literature. He is regarded as the pioneer of the **Aanchalik Upanyas (regional novel)** genre, depicting the language, culture, customs, and socio-economic realities of rural Bihar.

- **Maila Anchal:** Published in **1954**, Maila Anchal is Renu's most celebrated novel and a landmark in Hindi literature. Set in a village of north-eastern Bihar (Mithila region), it portrays rural life, poverty, caste relations, politics, healthcare, and the changes brought by India's independence. The novel established the **Aanchalik (regional)** style of writing in Hindi.
- **Parati Parikatha:** This novel presents the transformation of rural Bihar across generations. It explores the social, economic, and cultural changes affecting village life while vividly portraying the traditions and aspirations of rural communities.
- **Juloos:** Juloos deals with the settlement of **East Bengali refugees** in Bihar following the Partition and examines the resulting social tensions, cultural interaction, and challenges of rehabilitation.
- **Kitne Chaurahe:** This novel focuses on the **Indian freedom struggle**, depicting the sacrifices, ideals, and dilemmas faced by young revolutionaries who placed the nation's interests above their personal lives.

Q 45. Ans. B

Explanation:

- The **Bihar Cabinet** approved and launched the **Bihar Semiconductor Policy 2026** on **29 January 2026** with the objective of establishing Bihar as a major **semiconductor and electronics manufacturing hub in Eastern India**. The policy offers a range of incentives to attract large-scale investments in chip fabrication and electronics manufacturing.
- Key features of the policy include the **allotment of industrial land at a concessional rate of ₹1 per acre for every ₹100 crore invested**, along with **exemptions from stamp duty, registration charges, and land conversion fees**, besides providing **capital subsidies** on eligible project costs. Through this initiative, the state aims to attract **₹25,000 crore in investments** and create around **2 lakh direct and indirect employment opportunities**, with major industrial development planned in regions such as **Gaya** and **Muzaffarpur**.

Q 46. Ans. B

Explanation:

- According to the Economic Survey 2025–26, the Micro, Small, and Medium Enterprises (MSME) sector contributes approximately **31.1% to India's Gross Domestic Product (GDP)**.
- **Additional Facts:**
  - ✓ **Employment:** The sector remains India's second-largest employer (after agriculture), employing over **32.82 crore** people across more than 7.47 crore enterprises.
  - ✓ **Policy Support:** The Survey emphasised the role of the Self-Reliant India (SRI) Fund, which was launched to infuse ₹50,000 crore as equity funding and had invested over ₹15,442 crore across 682 MSMEs by late 2025.
  - ✓ **Formalisation Growth:** The Udyam Portal recorded a **14% increase** in total registrations in FY26 compared to FY25, bringing the total number of registered MSMEs to 7.47 crore.
  - ✓ **Credit Deficit Gap:** Despite robust growth, the Survey identified an **₹18.5 lakh crore** "credit gap" for micro-enterprises, which it cited as the single biggest bottleneck to achieving the 35%+ GDP contribution target.
  - ✓ **Interest Subvention:** The Survey noted that the **2% Interest Subvention Scheme** for MSMEs was extended through the end of FY27, specifically targeting exporters to mitigate the impact of high global interest rates.
  - ✓ **Energy Intensity:** A new metric introduced in the 2026 Survey: MSMEs accounted for **22% of total industrial energy consumption**, prompting the introduction of the "Green MSME" credit subsidy, which offers a 50-basis-point lower interest rate for firms adopting solar/energy-efficient technology.
  - ✓ **Geographic Concentration:** The Survey flagged that **62% of all registered MSMEs** are concentrated in just 6 states (Maharashtra, Tamil Nadu, Uttar Pradesh, Karnataka, Gujarat, and West Bengal), highlighting a severe regional imbalance in industrial growth.

Q 47. Ans. A

Explanation:

- **Statement 1 is incorrect:** The NMP focuses exclusively on unlocking the value of "brownfield" infrastructure assets (projects that are already built and operational). Crucially, it does not involve the outright sale or privatisation of these assets. Ownership strictly remains with the government, and the assets must be handed back to the public authority at the end of the designated concession period.
- **Statement 2 is correct:** The primary objective of the NMP is to generate upfront revenues or investments from private players using existing infrastructure. These proceeds are then explicitly meant to

be reinvested into creating new "greenfield" infrastructure projects under the massive National Infrastructure Pipeline (NIP).

- **Statement 3 is correct:** The framework identified core infrastructure sectors for monetisation. Roads (accounted for the largest share at 27%), Railways (25%), Power (15%), oil & gas pipelines, and Telecom are the primary sectors targeted to drive the bulk of the monetisation value.
- **Additional Facts:**
  - ✓ **Financial Target:** The original target for the National Monetisation Pipeline was to lease out ₹6 lakh crore worth of core infrastructure assets over a four-year period from FY2022 to FY2025.
  - ✓ **Key Investment Vehicles:** The NMP heavily relies on structured financial instruments to attract private and institutional capital. The most prominent among these are **InvITs** (Infrastructure Investment Trusts) and **REITs** (Real Estate Investment Trusts), alongside models like the **TOT** (Toll-Operate-Transfer) model used extensively by the National Highways Authority of India (NHAI).
  - ✓ **Brownfield vs. Greenfield:** In economics, a "brownfield" project refers to infrastructure that already exists (which the NMP targets). A "greenfield" project refers to building new infrastructure from scratch on unused land (which the NIP primarily funds).
  - ✓ **Nodal Agency:** The pipeline was developed by **NITI Aayog** in consultation with various infrastructure line ministries, under the mandate of the Ministry of Finance.

Q 48. Ans. C

Explanation:

- According to the Economic Survey 2025-26, India remained the undisputed global leader in remittance inflows, receiving a record **USD 135.4 billion** during the financial year 2024-25 (FY25).
- This massive inflow served as a crucial pillar in stabilising India's external account, helping to significantly offset the country's merchandise trade deficit.
- **The Survey highlighted a structural shift in remittance patterns:** while Gulf Cooperation Council (GCC) countries historically dominated inward remittances, advanced economies now contribute the largest share. This shift actively reflects the growing migration of highly skilled and professional Indian workers to these advanced economies.
- **Additional Facts:**
  - ✓ **Top Remittance Sources:** The United States is the top contributor to India's remittances, accounting for 27.7%, followed by the UAE (19.2%), the UK (10.8%), and Singapore (6.6%).
  - ✓ **Global Rankings:** According to World Bank data cited during this period, India's remittance inflows were nearly double that of the second-placed recipient country, Mexico ($68 billion), followed by China ($48 billion) and the Philippines ($40 billion).
  - ✓ **Forex Reserves:** The Economic Survey 2025-26 noted that India's foreign exchange reserves surged to **USD 701.4 billion** as of January 16, 2026. This robust buffer is sufficient to cover approximately 11 months of goods imports and 94% of India's outstanding external debt.
  - ✓ **Services Exports Record:** India's services exports touched an all-time high of **USD 387.6 billion** in FY25, growing by 13.6% year-on-year.
  - ✓ **FDI Dominance:** India retained its position as the largest recipient of gross Foreign Direct Investment (FDI) inflows in South Asia (surpassing peers like Vietnam and Indonesia) and emerged as the top global destination specifically for greenfield digital investments.

Q 49. Ans. C

Explanation:

- **Statement 1 is correct:** Financial repression is an umbrella term for government policies, including interest rate caps/ceilings, directed lending to the government via captive institutions (banks, pension funds), and capital controls restricting cross-border fund movement, that channel domestic funds toward the public sector rather than letting markets allocate them freely.
- **Statement 2 is correct:** By compelling banks and financial institutions to hold government debt (often below market interest rates), creating a "captive market", governments can borrow at artificially low cost, effectively financing fiscal deficits cheaply. Combined with inflation, this can even erode (liquidate) the real value of existing government debt, functioning as an implicit form of taxation.
- **Statement 3 is correct:** Because credit is directed by administrative fiat rather than market efficiency, financial repression tends to cause misallocation of

capital, discourage the development of competitive financial markets, and suppress financial innovation, a core finding of McKinnon and Shaw's original 1973 analysis (using South Korea as a case study), where such repression was shown to impair economic development until liberalised.

- **Statement 4 is incorrect:** Financial repression does the opposite; it is closely associated with "fiscal dominance," where monetary policy becomes subordinated to the government's need for cheap deficit financing, forcing central banks to keep interest rates artificially low rather than acting independently to target inflation or growth. This weakens, rather than strengthens, central bank independence, a dynamic explicitly noted in the historical example of the US Fed–Treasury conflict after WWII, which was resolved only when the 1951 US Fed–Treasury Accord ended fiscal dominance and restored a greater degree of Fed independence.

Q 50. Ans. A

Explanation:

- **Statement 1 is correct:** An angel investor is typically a high-net-worth individual who provides capital to early-stage startups or entrepreneurs, often investing their own personal funds (unlike venture capital firms, which manage pooled institutional/ limited-partner money).
- **Statement 2 is correct:** Angel investors generally provide funding in exchange for equity ownership (a stake) in the company, or sometimes via convertible debt that later converts into equity, aligning their returns with the company's future growth and eventual exit (via acquisition or IPO).
- **Statement 3 is incorrect:** Angel investors characteristically invest in companies at the very early/seed stage, often when the business is just an idea, a prototype, or in its infancy, well before the company is anywhere near being listed on a stock exchange. Investing in already-listed, publicly-traded companies falls under public market investing (buying shares on an exchange), which is a fundamentally different activity from angel investing. This statement essentially describes the opposite of what angel investors do.

Q 51. Ans. B

Explanation:

- **Bell-metal** is an alloy made primarily by combining **copper and tin**. It is a type of **bronze alloy** that contains a higher proportion of tin, which gives it hardness, strength, and good resonating properties. Due to these properties, bell-metal is traditionally used for making **bells, gongs, and other musical instruments**.
- Copper and tin together form **bronze**. A variety of bronze with a higher tin content is known as **bell-metal**, which produces a clear ringi is suitable for making sound andng bells.

Q 52. Ans. B

Explanation:

- **Xylem vessels** are specialised conducting elements that transport water and minerals efficiently from the roots to the aerial parts of the plant. They are the characteristic water-conducting elements of **angiosperms (flowering plants)** and provide much more efficient conduction than tracheids.
- Angiosperms commonly possess **well-developed xylem vessels**, which are the principal conducting elements for water transport. Along with vessels, their xylem also contains tracheids, xylem fibres, and xylem parenchyma. Hence, vessels are a characteristic feature of angiosperms.

Q 53. Ans. D

Explanation:

- The **proximal convoluted tubule (PCT)** is the major site of **selective reabsorption** in the nephron. Nearly **65–70% of the filtered sodium ions (Na)**, along with water, glucose, amino acids, and other useful substances, are **actively reabsorbed** from the filtrate back into the blood in this region. The active transport of sodium creates an osmotic gradient that also facilitates the reabsorption of water.
- The PCT is the **primary site of active sodium reabsorption**, accounting for the largest proportion of sodium reclaimed from the filtrate. Hence, it is considered the principal part of the nephron involved in active sodium reabsorption.

Q 54. Ans. D

Explanation:

- When potato tubers are exposed to sunlight, they develop a green colour due to chlorophyll formation. Along with this, the concentration of **solanine**, a toxic glycoalkaloid, also increases. Consumption of potatoes with high solanine content may cause nausea, vomiting, diarrhoea, and neurological symptoms.
- Oil extracted from the seeds of **Argemone mexicana** may accidentally adulterate mustard oil. The toxic alkaloids **sanguinarine** and **dihydrosanguinarine**

present in argemone oil can cause **epidemic dropsy**, characterised by swelling (oedema), cardiac problems, and glaucoma.

- Groundnuts stored under warm and humid conditions may become infected with **Aspergillus flavus**, which produces **aflatoxins**. These toxins are highly poisonous and carcinogenic. Contaminated seeds often develop an unpleasant or bitter taste and should not be consumed.
- **Cucurbitacins** are bitter compounds naturally found in plants of the **Cucurbitaceae** family. However, **they are not present in all cucumbers in harmful amounts**. Most cultivated cucumbers have been bred to contain **very low or negligible levels of cucurbitacins**. Only occasionally, due to genetic factors or environmental stress, some cucumbers become unusually bitter because of increased cucurbitacin content. Therefore, it is incorrect to state that cucurbitacin is present in **all** cucumbers.

Q 55. Ans. A

Explanation:

- **Assertion (A) is true:** A single **displacement reaction** is a reaction in which a **more reactive element displaces a less reactive element from its compound**. Such reactions always involve the **transfer of electrons**, making them **redox (oxidation–reduction) reactions**. In a displacement reaction, one element **loses electrons (oxidation)** while another **gains electrons (reduction)**. Since oxidation and reduction occur simultaneously, every single displacement reaction is a **redox reaction**.
- **Reason (R) is true:** Electron transfer is the fundamental feature of a displacement reaction. The more reactive element donates electrons, while the less reactive element accepts them. This electron transfer is exactly what constitutes oxidation and reduction.
- **Since the Reason correctly explains why every single displacement reaction is a redox reaction, it is the correct explanation of the Assertion. Therefore, the correct answer is: A. Both (A) and (R) are true, and (R) is the correct explanation of (A).**

Q 56. Ans. C

Explanation:

- **Highly reactive metals** such as **potassium, sodium, calcium, magnesium, and aluminium** cannot be extracted by chemical reducing agents like carbon because they have a greater affinity for oxygen. Therefore, these metals are extracted by the **electrolytic reduction of their molten compounds**.
- Sodium is a highly reactive metal. It is extracted by the **electrolysis of molten sodium chloride (Downs process)** because carbon cannot reduce its oxide.
- Magnesium is also highly reactive. It is obtained by the **electrolysis of molten magnesium chloride** in industrial processes.
- Aluminium is extracted by the **electrolysis of molten alumina** (Al_2O_3) dissolved in molten cryolite. This process is known as the **Hall–Héroult process**.
- **Zinc is moderately reactive and is not extracted by electrolytic reduction. Its oxide (ZnO) is reduced using carbon (coke) during the extraction process:**

Q 57. Ans. C

Explanation:

- When **zinc** is placed in a **copper sulphate solution**, zinc being more reactive displaces copper from the solution. As a result, **reddish-brown copper metal gets deposited** on the zinc strip, and the blue colour of the solution gradually fades due to the formation of zinc sulphate.
- Aqueous **copper sulphate solution** is characteristically **blue** because of the presence of hydrated **Cu^2 ions**.
- When **silver nitrate solution** is mixed with **sodium chloride solution**, a **white precipitate of silver chloride (AgCl)** is formed.
- When **magnesium ribbon burns in air**, it burns with a dazzling white flame and forms a **white ash of magnesium oxide (MgO)**.

Q 58. Ans. D

Explanation:

- **Statement 1 is correct:** Chlorine has the electronic configuration **2, 8, 7**. It requires **one electron** to complete its octet and therefore has a **valency of one**. This is why chlorine forms compounds such as **HCl** and **NaCl**, where it combines with one atom of hydrogen or sodium.
- **Statement 2 is correct:** Oxygen has the electronic configuration **2, 6**. It requires **two electrons** to complete its octet and therefore has a **valency of two**. Hence, oxygen forms compounds such as H_2O, where it combines with two hydrogen atoms.
- **Statement 3 is correct:** Nitrogen has the electronic configuration **2, 5**. It requires **three electrons** to complete its octet and commonly exhibits a **valency of three**, as seen in compounds like NH_3 **(ammonia)**.

Q 59. Ans. B

Explanation:

- **The magnetic field produced inside a long solenoid is directly proportional to the number of turns per unit length and the current flowing through it. The formula for the magnetic field inside a solenoid is:**
  - ✓ $B = \mu_0 n I$
  - ✓ where **B** is the magnetic field strength, is the permeability of free space, **n** is the number of turns per unit length, and **I** is the current through the solenoid.
- The number of turns per unit length is given by **n = N/L**, where **N** represents the total number of turns and **L** represents the length of the solenoid.
- In the given question, both solenoids P and Q have the same total number of turns and carry the same current. Therefore, the magnetic field strength depends only on the length of the solenoid. Solenoid P is twice as long as solenoid Q.
- Let the length of solenoid Q be **L**. Then the length of solenoid P will be **2L**.
  - ✓ **For solenoid P:**
  - ✓ $n_p = N/2L$
  - ✓ **For solenoid Q:**
  - ✓ $n_p = N/L$ Therefore,
  - ✓ $B_p : B_p = n_p : n_p$
  - ✓ $= (N/2L) : (N/L)$
  - ✓ **= 1 : 2**
- Thus, the magnetic field strength inside solenoid P is half of that inside solenoid Q.

Q 60. 60. Ans. A

Explanation:

- **Vermicomposting** is a method of preparing compost with the help of **earthworms**. Earthworms feed on organic waste and convert it into nutrient-rich manure called **vermicompost**. They help in decomposition by increasing microbial activity and improving the fertility of the soil.
- Eisenia foetida (commonly known as **red wiggler or red earthworm**) is one of the most commonly used earthworms for **vermicomposting in India**. It has a high rate of reproduction, survives well in organic waste, and efficiently converts biodegradable material into compost.

Q 61. Ans. B

Explanation:

- **Sieve tube elements** are the principal conducting cells of the **phloem**. During their maturation, **they lose their nucleus** and several other organelles to provide an unobstructed pathway for the translocation of food (mainly sucrose). However, they remain **living cells** because their metabolic activities are controlled by the adjacent **companion cells**, which retain a prominent nucleus and are connected to the sieve tube elements through plasmodesmata.
- Sieve tube elements are **living but enucleate (without a nucleus) at maturity**. Their cytoplasm remains functional due to the close association with companion cells. This is a unique feature of angiosperm phloem and allows efficient transport of food.

Q 62. Ans. C

Explanation:

- **B-lymphocytes (B-cells)** are the white blood cells primarily responsible for **humoral immunity**. When a foreign antigen enters the body, B-lymphocytes become activated and differentiate into **plasma cells**, which secrete **specific antibodies (immunoglobulins)** against that antigen. These antibodies neutralise pathogens, promote their destruction, and provide long-term immunity through the formation of **memory B-cells**.
- B-lymphocytes are responsible for **antibody-mediated (humoral) immunity**. After activation, they transform into plasma cells, which synthesise and release antibodies into the bloodstream.

Q 63. Ans. B

Explanation:

- **Secondary growth** is the increase in the **girth (thickness)** of stems and roots that occurs mainly in **dicotyledonous plants and gymnosperms**. This growth is brought about by the activity of **lateral meristems**, chiefly the **vascular cambium** and **cork cambium**.
- The vascular cambium is a **lateral meristem** located between the xylem and phloem. During secondary growth, it divides actively to produce **secondary xylem (wood) towards the inside** and **secondary phloem towards the outside**. This continuous production of secondary vascular tissues increases the diameter of the stem, making the vascular cambium the principal tissue responsible for secondary growth in dicot stems.

Q 64. Ans. D

Explanation:

- **Pair 1 is correct: Alum** (commonly potassium aluminium sulphate, $KAl(SO_4)_2 \cdot 12H_2O$) is widely used in **water purification** as a **coagulant**.

It causes suspended impurities to coagulate into larger particles (flocs), which settle down and can be removed by filtration. This process is commonly employed in municipal water treatment plants.

- **Pair 2 is correct:** Ammonium phosphates, such as **diammonium phosphate (DAP)** and **monoammonium phosphate (MAP)**, are important **chemical fertilisers**. They supply both **nitrogen (N)** and **phosphorus (P)**, two essential nutrients required for healthy plant growth and crop productivity.
- **Pair 3 is correct:** Anhydrous **aluminium chloride** ($AlCl_3$) is an important **Lewis acid catalyst** used extensively in **organic synthesis**, particularly in **Friedel–Crafts alkylation and acylation reactions** for the preparation of aromatic compounds.
- **Pair 4 is correct:** Graphite is a good **conductor of electricity** due to the presence of delocalized electrons between its carbon layers. Because of its electrical conductivity, high melting point, and chemical inertness, it is widely used for making **electrodes** in batteries, electrolysis, and electric furnaces.

Q 65. Ans. B

Explanation:

- **Statement 1 is incorrect: Sodium carbonate** (Na_2CO_3), commonly known as **washing soda**, is **not used as a rat poison**.it is mainly used:
  - ✓ In the manufacture of glass, soap and paper.
  - ✓ For removing permanent hardness of water.
  - ✓ As a cleaning agent in homes and industries.
  - ✓ Rat poisons generally contain compounds such as **zinc phosphide**, **warfarin**, or **bromadiolone**, not sodium carbonate.
- **Statement 2 is correct: Potassium bromide (KBr)** is used in the **photographic industry** for the preparation of **silver bromide (AgBr)**, a light-sensitive compound used in photographic films and papers. KBr also acts as a restrainer in photographic developers to reduce fogging.
- **Statement 3 is correct:** Metallic **sodium** is used in **vapour lamps**, which are widely employed for **street and highway lighting** because they produce bright yellow light with high luminous efficiency. Although the reflector itself is not made of sodium, sodium is the essential element used in these road lamps.

Q 66. Ans. B

Explanation:

- **Natural rubber** is a naturally occurring polymer obtained from the latex of the rubber tree (Hevea brasiliensis). It is composed of repeating units of **isoprene (2-methyl-1,3-butadiene)**.
- Natural rubber is formed by the **1,4-addition polymerisation** of **cis-isoprene** molecules. Hence, it is also called **cis-1,4-polyisoprene**. The **cis configuration** gives natural rubber its characteristic properties such as **high elasticity, flexibility, and resilience**.
- Isoprene (2-methyl-1,3-butadiene) is the **monomer of natural rubber**. Thousands of isoprene molecules join together through addition polymerisation to form **cis-1,4-polyisoprene**, which constitutes natural rubber.

Q 67. Ans. A

Explanation:

- **Ethyne (acetylene, C_2H_2)** is the simplest member of the **alkyne** family of hydrocarbons. Alkynes are **unsaturated hydrocarbons** containing at least one **carbon–carbon triple bond (C≡C)**.
- Ethyne has the structure **H–C≡C–H**, in which the two carbon atoms are connected by a **triple covalent bond**. Due to this triple bond, it is an **unsaturated hydrocarbon** and is chemically more reactive than alkanes. Ethyne is **highly flammable** and burns with a luminous flame in air. When burned with pure oxygen, it produces an extremely hot flame (about **3300°C**), which is used in **oxy-acetylene welding and metal cutting**.

Q 68. Ans. B

Explanation:

- **Assertion (A) is true:** elasticity is the property of a material by virtue of which it regains its original shape and size after the deforming force is removed, provided the elastic limit is not exceeded. A **perfectly elastic body** completely regains its original dimensions after the removal of the external force. In practice, no material is perfectly elastic, but materials such as quartz and phosphor bronze exhibit nearly perfect elastic behaviour within their elastic limits.
- **Reason (R) is true: Within the elastic limit, stress is directly proportional to strain.** This statement is **Hooke's Law**, which states that within the elastic limit. This law describes the **linear relationship between stress and strain** during elastic deformation.
- **Although Hooke's Law is true, it does not explain why a body regains its original shape. The recovery of shape is due to the property of elasticity of the material. Hooke's Law merely**

states the mathematical relationship between stress and strain within the elastic limit. Thus, it describes elastic behaviour quantitatively but is not the fundamental reason for the body's ability to recover its original shape. Therefore, the correct answer is: B. Both (A) and (R) are true, but (R) is not the correct explanation of (A).

Q 69. Ans. A

Explanation:

- **Assertion (A) is true:** a **convex lens (converging lens)** can produce **both real and virtual images**, depending on the position of the object with respect to its principal focus. A convex lens forms different types of images depending on where the object is placed:
  - ✓ **Object beyond the focal length (F):** The lens forms a **real and inverted image** on the opposite side of the lens.
  - ✓ **Object at infinity:** A highly diminished real image is formed at the focus.
  - ✓ **Object between the optical centre and the focal point (within the focal length):** The lens forms a **virtual, erect, and magnified image** on the same side of the lens.
  - ✓ Thus, a convex lens is capable of producing **both real and virtual images**.
- **Reason (R) is true:** the **nature, position, and size of the image formed by a convex lens depend entirely on the object's position relative to the focal length.** If the object is beyond the focal point, the image is real; if the object is within the focal length, the image is virtual. Thus, the ability of a convex lens to produce both real and virtual images is **directly explained by the object's position with respect to the focal length.**
- **Since both the Assertion and the Reason are true, and the Reason correctly explains the Assertion, the correct answer is: A. Both (A) and (R) are true, and (R) is the correct explanation of (A).**

Q 70. Ans. D

Explanation:

- The cyclist moves in two stages. In the first stage, he starts from rest (**initial velocity, u = 0**) and accelerates uniformly at **1.5 m/s²** for **20 seconds.** The speed attained after 20 seconds is calculated using the equation **v = u + at**, which gives **v = 0 + (1.5 × 20) = 30 m/s.** The distance covered during this uniformly accelerated motion is calculated using the equation **s = ut + ½at²**, giving **s = 0 + ½ × 1.5 × (20)² = 300 m.**
- **In the second stage, the cyclist continues to move with the attained constant speed of 30 m/s for 40 seconds. The distance covered during uniform motion is calculated using s = vt = 30 × 40 = 1200 m. Therefore, the total distance travelled by the cyclist is 300 + 1200 = 1500 m. Hence, the correct answer is: D. 1500 m.**

Q 71. Ans. B

Explanation:

- When a force acts at an angle to the direction of displacement, only the component of the force acting along the direction of displacement performs work. The work done is calculated using the formula **W = Fs cos** , where **F** is the applied force, **s** is the displacement, and is the angle between the force and the displacement.
- **In this question, the applied force is 80 N, the displacement is 15 m, and the angle between them is 60°. Since cos 60° = 1/2, the work done is calculated as W = 80 × 15 × 1/2 = 600 J. The component of the force perpendicular to the displacement does not contribute to the work done. Hence, the correct answer is: B. 600 J.**

Q 72. Ans. C

Explanation:

- The image formed by a spherical mirror is calculated using the **mirror formula, 1/f = 1/v + 1/u.** For a concave mirror, the focal length and object distance are taken as negative according to the Cartesian sign convention. Here, the object distance **u = −15 cm** and the focal length **f = −10 cm.**
- **Substituting these values into the mirror formula:**
  - ✓ **1/−10 = 1/v + 1/−15**
  - ✓ **1/v = −1/10 + 1/15 = (−3 + 2)/30 = −1/30**
  - ✓ Therefore, **v = −30 cm**
- The negative sign indicates that the image is formed **in front of the mirror**, which means it is a **real and inverted image.** Since the options provide only the magnitude of the image distance, the correct answer is **30 cm.**

Q 73. Ans. B

Explanation:

- Porifera (sponges) are the simplest multicellular animals and exhibit a cellular level of organisation. In these animals, the cells are loosely aggregated and perform different functions, but they are not organised into true tissues or organs. Different types of cells, such as choanocytes, pinacocytes, and

amoebocytes, work together to carry out various life processes, yet the body lacks true tissue formation.

- **Cnidaria** is incorrect because members of this phylum exhibit a **tissue-level organisation**. They possess two distinct germ layers (diploblastic body plan), and their cells are organised into tissues that perform specialised functions.
- **Platyhelminthes** is incorrect because they exhibit an **organ level of organisation**. These animals are triploblastic, bilaterally symmetrical, and possess well-developed organs such as the digestive and excretory systems.
- **Annelida** is also incorrect because annelids possess an **organ-system level of organisation**, where different organs work together as complete systems such as the digestive, circulatory, nervous, and excretory systems.

Q 74. Ans. B

Explanation:

- **Cytokinins** are plant growth regulators that promote **cell division (cytokinesis)** and help **delay the ageing (senescence) of leaves**. They maintain chlorophyll content, prolong the functional life of leaves, and are therefore known as **anti-senescence hormones**. Hence, the pair **"Cytokinin — Delays senescence"** is correctly matched.
- **Auxin — Promotes stomatal closure** is **incorrect** because auxins mainly promote **cell elongation, apical dominance, root initiation, and fruit development**. Stomatal closure is primarily regulated by **abscisic acid (ABA)**, especially under conditions of water stress.
- **Ethylene — Promotes apical dominance** is **incorrect** because **apical dominance** is mainly controlled by **auxins** produced at the shoot apex. Ethylene is involved in **fruit ripening, leaf and flower senescence, and abscission**, rather than maintaining apical dominance.
- **ABA — Stimulates seed germination** is **incorrect** because **abscisic acid (ABA)** generally **induces seed dormancy** and inhibits germination. Seed germination is mainly promoted by **gibberellins**, which break dormancy and stimulate enzyme production required for germination.

Q 75. Ans. D

Explanation:

- **Photosynthesis occurs in two major stages:** the **light reaction** and the **dark reaction (Calvin cycle)**. During the light reaction, which takes place in the **thylakoid membranes** of chloroplasts, light energy is absorbed by chlorophyll and converted into chemical energy.
- The major products of the **light reaction** are **ATP**, **NADPH**, and **oxygen** (O_2). ATP and NADPH are energy-rich molecules that provide the energy and reducing power required for the synthesis of carbohydrates during the Calvin cycle. Oxygen is released as a by-product due to the **photolysis (splitting) of water**.
- **Glucose** is **not** a direct product of the light reaction. It is synthesised during the **Calvin cycle (light-independent reaction)** in the stroma of the chloroplast using **carbon dioxide, ATP, and NADPH** produced during the light reaction.

Q 76. Ans. C

Explanation:

- **Mycoplasma** are the **smallest known living cells** that are capable of **independent existence**. They are unique among prokaryotes because they **completely lack a cell wall**. Instead, they possess a flexible plasma membrane containing sterols, which provides structural support and allows them to survive without a rigid cell wall. This absence of a cell wall also makes them naturally resistant to antibiotics such as penicillin, which act by inhibiting cell wall synthesis.
- **Statement 1 is correct:** Mycoplasma are the only prokaryotic organisms that naturally **lack a cell wall** throughout their life cycle. Despite this, they survive and reproduce independently due to their specialised plasma membrane.
- **Statement 2 is correct: Mycoplasma are the smallest living cells known** and are capable of carrying out all essential life processes independently. Their size is generally about **0.1–0.3 m**, making them smaller than most bacteria.

Q 77. Ans. A

Explanation:

- **Statement 1 is correct:** The **Law of Segregation** states that the two alleles of a gene separate from each other during **gamete formation**, so that each gamete receives only one allele. As a result, gametes remain **genetically pure** with respect to a particular character. Therefore, this law is also known as the **Law of Purity of Gametes**.
- **Statement 2 is incorrect:** The **Law of Independent Assortment** states that alleles of different genes assort independently during gamete formation.

However, this law is valid only for **genes located on different chromosomes** or **genes that are sufficiently far apart on the same chromosome** so that recombination occurs frequently. **Completely linked genes**, which are located very close together on the same chromosome, are inherited together and **do not assort independently**. Therefore, Mendel's Law of Independent Assortment **does not apply to completely linked genes**

Q 78. Ans. C

Explanation:

Venera-D is Russia's upcoming flagship mission to **Venus**, with a planned launch window between **2034 and 2036**, marking the revival of the historic **Soviet-era Venera programme**. The mission aims to carry out a comprehensive investigation of Venus's **atmosphere, surface, and plasma environment**. It comprises **three major components**: an **orbiter**, which will conduct radar mapping, capture high-resolution images, and relay scientific data; a **lander**, designed to survive for about **three hours** on Venus's extreme surface while studying its **soil, atmospheric pressure, and seismic activity**; and a **balloon probe**, which will float through the planet's upper atmosphere to analyze **winds, temperature, and chemical composition**, while also investigating the possibility of **microbial life in the Venusian clouds**.

Q 79. Ans. B

Explanation:

Group Captain Shubhanshu Shukla safely returned to Earth on **15 July 2025** after completing a historic **18-day mission aboard the International Space Station (ISS)** as part of **Axiom Mission 4 (Ax-4)**. Launched on **25 June 2025**, the mission marked the **first time an Indian astronaut travelled to the ISS**. During his stay, Shukla carried out **more than 60 scientific experiments**, many of which are expected to support **ISRO's Gaganyaan human spaceflight programme**. Following post-mission rehabilitation and training in the **United States**, he returned to **India on 17 August 2025**, where he participated in national events and met **Prime Minister Narendra Modi** to share insights from the mission.

Q 80. Ans. C

Explanation:

- **Sarvam AI**, a Bengaluru-based artificial intelligence startup, officially became **India's 130th unicorn** after raising **US$234 million** in the initial close of its **Series B funding round**, taking its valuation to approximately **US$1.5 billion**. The funding round was led by **HCLTech**, which invested **US$150 million** and acquired an estimated **10.5% stake** in the company. Other prominent investors included **Bessemer Venture Partners, Khosla Ventures, and Peak XV Partners**, while the company aims to raise a total of **US$300 million** in the complete funding round.
- Sarvam AI focuses on building a **sovereign AI ecosystem** tailored to India's needs rather than relying on foreign AI platforms. Its core work includes developing **indigenous Large Language Models (LLMs)** such as **Sarvam-30B** and **Sarvam-105B**, creating **AI-powered voice agents** capable of natural conversations in multiple Indian languages, and building AI infrastructure optimised for regional use cases. The company's technologies are already being deployed in **government and enterprise projects**, including document digitisation, multilingual speech processing, and data collection initiatives for rural farmers, demonstrating the practical application of AI at scale across India.

Q 81. Ans. A

Explanation:

- **Statement 1 is correct:** Early human settlements were heavily concentrated near river valleys (like the Narmada, Soan, and Tungabhadra) to ensure access to drinking water, game animals, and raw stones for tool-making.
- **Statement 2 is correct:** Bhimbetka (Raisen district) and Adamgarh (Narmadapuram district) are critical prehistoric sites in Madhya Pradesh, yielding continuous habitation evidence from the Palaeolithic to the Mesolithic eras.
- **Statement 3 is incorrect:** Attirampakkam is a prominent Lower Palaeolithic site located near Chennai in **Tamil Nadu**, not Kerala. It is globally famous for its Acheulean handaxes.
- **Additional Facts:**
  - ✓ **First Discovery:** Robert Bruce Foote, a British geologist, discovered the first Palaeolithic tool in India (a handaxe at Pallavaram, Tamil Nadu) in 1863.
  - ✓ **Narmada Man:** Hathnora (Madhya Pradesh) is the site where the earliest hominid fossil in India (Homo erectus, dubbed 'Narmada Man') was discovered.
  - ✓ **Hunsgi Valley (Karnataka):** A unique Lower

Palaeolithic site where early humans primarily manufactured tools out of local limestone, rather than the usual quartzite.

- ✓ **Earliest Site:** Bori in Maharashtra is considered one of the earliest dated Lower Palaeolithic sites in the Indian subcontinent.
- ✓ **Tool Typology Phase Shift:**
  - **Lower Palaeolithic:** Handaxes, cleavers, choppers.
  - **Middle Palaeolithic:** Flake-based tools (scrapers).
  - **Upper Palaeolithic:** Blade and burin tools (coinciding with the emergence of Homo sapiens and a drier climate).

Q 82. Ans. B

Explanation:

- **Langhnaj (Gujarat):** This is a classic **Mesolithic** type-site. It is famous for yielding microliths, animal bones, and some of the earliest evidence of deliberate human burials in the Indian subcontinent.
- **Burzahom:** A strictly **Neolithic** site in Kashmir, famous for pit dwellings and burying dogs with their masters.
- **Lothal & Kalibangan (Incorrect):** Both are prominent Bronze Age (Chalcolithic) sites belonging to the Indus Valley (Harappan) Civilization.
- **Additional Facts:**
  - ✓ **Microliths:** The defining technology of the Mesolithic Age. These are tiny, geometric stone tools (like blades, triangles, and lunates) ranging from 1 to 5 cm in length.
  - ✓ **Bagor (Rajasthan):** The largest and best-documented Mesolithic site in India, located on the Kothari River. It provides the earliest evidence of animal domestication.
  - ✓ **Adamgarh (Madhya Pradesh):** Along with Bagor, this site provides the earliest evidence for the domestication of animals (primarily dogs and cattle).
  - ✓ **Sarai Nahar Rai & Mahadaha (Uttar Pradesh):** Famous for bone tools, ornaments, and early evidence of **double burials** (men and women buried together).
  - ✓ **Chopani Mando (Uttar Pradesh):** Provides crucial evidence of a continuous sequence from the Mesolithic to the Neolithic transition, including the earliest use of handmade pottery (pottery is generally absent in most Mesolithic sites).

Q 83. Ans. A

Explanation:

- **Steatite (soapstone)** was the most commonly used material for Harappan beads and seals. Because it is an extremely soft stone, it was easy to cut, shape, and carve into intricate designs.
- **The Technique:** Harappans also mastered a technique of making a paste from micro-steatite powder, molding it into diverse shapes, and then firing it at high temperatures to harden it.
- **Additional Facts:**
  - ✓ **Factory Sites: Chanhudaro** (in Sindh, Pakistan) and **Lothal** (in Gujarat, India) were the most prominent, specialized centers for bead-making, evidenced by the discovery of "bead factories" and specialized drill bits.
  - ✓ **Carnelian (The Red Stone):** A highly prized material for beads. The yellowish raw material and the finished beads were fired at different stages of production to obtain the signature reddish color.
  - ✓ **Lapis Lazuli (The Blue Stone):** A prestige material imported from **Shortughai** (in modern-day Afghanistan), used for high-status beads.
  - ✓ **Other Materials:** Besides steatite, Harappans used agate, jasper, crystal, quartz, faience (an artificially produced material), terracotta, gold, and copper.
  - ✓ **Bead Shapes:** They were highly varied, disc-shaped, cylindrical, spherical, barrel-shaped, and segmented. Some were etched or painted with designs.

Q 84. Ans. B

Explanation:

- In the Rigvedic (Early Vedic) period, cattle were the primary measure of wealth. The term **Gavishthi** literally translates to "search for cows," but it was practically used to denote a war, battle, or raid fought to capture cattle from rival tribes.
- Sacrificial rituals were known as Yajnas, tribal assemblies were Sabhas or Samitis, and land grants were virtually unknown during this period because land was not considered private wealth.
- **Additional Facts:**
  - ✓ **Gomat:** A wealthy person (literally, one who owns many cows).
  - ✓ **Duhitri:** The term for daughter (literally, one who milks the cow).

- ✓ **Aghanya:** The cow was accorded the status of Aghanya (that which is not to be killed or slaughtered).
- ✓ **Raja / Gopa:** The tribal chief's primary duty was to protect the tribe's cattle (Gopa means protector of cows), rather than ruling over a fixed, demarcated territory.
- ✓ **Pastoral Economy:** The Early Vedic economy was predominantly pastoral. Agriculture was of secondary importance, and private, individual ownership of land did not exist.

Q 85. Ans. C

Explanation:

- **Pair 1 is correct:** After Ashoka's death (c. 232 BCE), the vast Mauryan Empire was partitioned. **Kunala** (Ashoka's son) took control of the **western part** of the empire, with its administrative center at Ujjain.
- **Pair 2 is correct: Dasaratha** (Ashoka's grandson) retained control over the **eastern part** of the empire, ruling from the imperial capital of Pataliputra.
- **Pair 3 is correct:** Dasaratha ruled for about eight years and was succeeded by **Samprati** (Kunala's son). Samprati is credited with reuniting the eastern and western halves of the empire.
- **Additional Facts:**
  - ✓ **Dasaratha & The Ajivikas:** Dasaratha is the only Later Mauryan emperor known to have issued epigraphic inscriptions. He dedicated the rock-cut caves at Nagarjuni Hills (near Barabar, Bihar) to the Ajivika sect and continued the use of Ashoka's royal title, Devanampiya.
  - ✓ **The "Jain Ashoka":** Samprati was a devout patron of Jainism. Converted by the Jain monk Suhastin, he built thousands of Jain temples across India, playing a role in propagating Jainism that closely mirrors Ashoka's role in spreading Buddhism.
  - ✓ **The Fall:** The Mauryan dynasty officially ended in 184 BCE when its last ruler, **Brihadratha**, was assassinated during a military parade by his own commander-in-chief, Pushyamitra Shunga, who then established the Shunga Dynasty.

Q 86. Ans. A

Explanation:

- **Ustad Mansur** was the leading painter in Emperor Jahangir's court, celebrated for his incredibly detailed and scientifically accurate illustrations of plants and animals. In recognition of his mastery, Jahangir bestowed upon him the title **Nadir-ul-Asr** (Wonder of the Age).
- **Additional Facts:**
  - ✓ **Mansur's Masterpieces:** He is globally famous for painting the Siberian Crane and the now-extinct Mauritius Dodo.
  - ✓ **Abul Hasan:** Another chief painter in Jahangir's court, who specialized in portraiture and court scenes. He was honored with the title **Nadir-uz-Zaman** (Wonder of the Times).
  - ✓ **Bishandas:** Praised by Jahangir as the undisputed master of portraits, he was sent as an envoy to Persia specifically to paint the portrait of the Safavid King, Shah Abbas.
  - ✓ **Daswanth & Basawan:** These were prominent pioneer painters during the earlier reign of **Akbar**, known for illustrating the Razmnama (the Persian translation of the Mahabharata).
  - ✓ **Zenith of Art:** While Shah Jahan's reign was the golden age of architecture, **Jahangir's** reign is universally recognized as the zenith (golden age) of Mughal painting, characterized by a shift towards naturalism and individual portraits rather than just manuscript illustrations.

Q 87. Ans. A

Explanation:

- During the Later Vedic Period, regular coinage in the modern sense had not yet developed, but metallic pieces of specific weights were widely used as a medium of exchange. **Nishka** (gold), **Satamana** (silver or gold), and **Krishnala** (a standard unit of weight based on a berry) served this purpose.
- **Additional Facts:**
  - ✓ **Evolution of Nishka:** In the Early Vedic (Rigvedic) period, Nishka originally referred to a gold necklace or jewelry. By the Later Vedic period, it evolved into an accepted unit of currency.
  - ✓ **Punch-Marked Coins:** The earliest actual minted coins in India, appearing around the 6th century BCE (Mahajanapada era). Made mostly of silver (and some copper), they bore punched symbols like trees, hills, and animals, but no inscriptions.
  - ✓ **Indo-Greeks:** They were the first rulers in India to issue coins that could be definitively attributed to specific kings (bearing portraits and names), and the first to issue **gold coins**.
  - ✓ **Gupta Coinage:** The Guptas issued the largest

number of gold coins in ancient India, called **Dinaras**. Their silver coins, issued after defeating the Shakas, were called **Rupakas**.

- ✓ **Satavahanas:** Their base-metal coins were primarily made of copper, lead, and potin alloy (an alloy of copper, tin, lead, and sometimes zinc)
- ✓ **Delhi Sultanate:** Iltutmish introduced the standard Arabic coinage system: the silver **Tanka** and the copper **Jital**.

Q 88. Ans. B

Explanation:

- **Pair 1 is incorrect:** The Sandhi-vigrahika was the **Minister of War and Peace** (foreign affairs), a post created during the Gupta period and continued by Harsha.
- **Pair 2 is incorrect:** The Prathama-kulika was the **Chief Craftsman** or the head of the artisans' guild.
- **Pair 3 is correct:** The Maha-danda-nayaka was the **Chief Judicial Officer** or Chief Magistrate.
- **Additional Facts:**
  - ✓ **Nagara-shreshthi:** The Chief Banker or chief merchant of the city (representing the urban commercial interests).
  - ✓ **Sarthavaha:** The leader of the merchant caravans.
  - ✓ **Kumaramatya:** The most important cadre of high executive officers (acting much like the modern IAS). Many central ministers were chosen from this rank.
  - ✓ **Uparika (or Uparika-Maharaja):** The Governor of a province (Bhukti).
  - ✓ **Ayuktas:** Officers entrusted with the restoration of the wealth of kings conquered in war and general administrative duties.
  - ✓ **Harsha's Specific Titles:** During Harsha's reign, the cavalry officers were known as Brihadashvamara, and the infantry officers were called Baladhikritas or Mahabaladhikritas.

Q 89. Ans. D

Explanation:

- The **Hamzanama** (Epic of Amir Hamza) was the foundational project of the Mughal school of painting, commissioned by Akbar in the 1560s.
- **The Transition:** Taking 15 years to complete, the project was directed by two Persian master artists (Mir Sayyid Ali and Abd al-Samad) who supervised a workshop of over 100 Indian painters (both Hindu and Muslim). The daily collaboration between Persian supervisors and Indian artists organically fused Persian techniques with Indian aesthetics, birthing the distinct "Mughal" style.
- **Additional Facts:**
  - ✓ **Massive Scale:** The Hamzanama was unprecedented in scale, consisting of **1,400 large illustrations** painted on cotton cloth (unlike later manuscripts painted on paper).
  - ✓ **Tutinama (Tales of a Parrot):** Another very early manuscript from Akbar's reign. It is often cited as the first work where the Indian influence (brighter colors, distinct facial profiles) began creeping into the Persian framework.
  - ✓ **Razmnama:** The Persian translation of the Mahabharata. It was the most important project of Akbar's later reign, illustrated heavily by prominent Indian artists like **Daswanth** (who tragically took his own life) and **Basawan**.
  - ✓ **European Influence:** It was during Akbar's reign that Portuguese Jesuits introduced European paintings and engravings (especially Christian themes) to the Mughal court, bringing the concepts of **perspective (foreshortening)** and **volumetric shading** into Mughal art.

Q 90. Ans. D

Explanation:

- Built in the 1560s, Humayun's Tomb was a watershed moment in Mughal architecture. It is widely recognized as the first true "garden-tomb" on the Indian subcontinent, introducing the formal Persian **Charbagh** (four-quadrant garden) layout.
- It is also the first Indian structure to utilize the Persian **double dome** concept on a massive, structural scale. It is often referred to as the precursor to the Taj Mahal.
- **Additional Facts:**
  - ✓ **Double Dome Utility:** It consists of two structural layers. The tall outer dome provides a grand, imposing exterior height, while the lower inner dome ensures the interior ceiling remains proportionate and acoustically sound.
  - ✓ **Itimad-ud-Daula's Tomb (Agra):** Built by Nur Jahan for her father, it is crucial for exams as it marks the architectural transition from red sandstone to pure white marble. It is also the first Mughal structure to extensively feature **Pietra Dura** (floral inlay work using semi-precious stones).
  - ✓ **Buland Darwaza (Fatehpur Sikri):** Built by Akbar specifically to commemorate his military victory over Gujarat.

- ✓ **Shah Jahan's Reign:** Marked the zenith of Mughal architecture, shifting heavily from Akbar's preferred red sandstone to pristine white marble (e.g., the Taj Mahal, which represents the ultimate perfection of the Charbagh and double dome concepts).

Q 91. Ans. B

Explanation:

- Gopal Krishna Gokhale was the preeminent leader of the **Moderate** faction within the Indian National Congress.
- He firmly believed in constitutional agitation, peaceful protests, and securing gradual political reforms through dialogue and petitions to the British government. He stood in direct contrast to the aggressive, direct-action methods of the Extremist faction (led by Lal-Bal-Pal).
- **Additional Facts:**
  - ✓ **Political Guru:** Gokhale is famously known as the political guru of **Mahatma Gandhi**, who later wrote a book titled Gokhale, My Political Guru. In turn, Gokhale considered M.G. Ranade as his own mentor.
  - ✓ **Servants of India Society:** He founded this organization in 1905 in Pune to train Indians as national missionaries and to promote education and social reform.
  - ✓ **1905 Benaras Session:** Gokhale presided over this crucial INC session. While the Congress formally supported the Swadeshi movement against the partition of Bengal here, ideological cracks widened between Moderates and Extremists over how far the movement should spread.
  - ✓ **Surat Split (1907):** The escalating rivalry between Gokhale's Moderates and Bal Gangadhar Tilak's Extremists ultimately culminated in the historic splitting of the Congress at the Surat session.
  - ✓ **Imperial Legislative Council:** He was a prominent member of the Imperial Legislative Council, playing a key role in shaping the Morley-Minto Reforms of 1909.
  - ✓ **Hitavada:** He launched the English weekly newspaper The Hitavada (The People's Paper) in 1911.

Q 92. Ans. C

Explanation:

- The **Indian Councils Act of 1909** (Morley-Minto Reforms) retained the absolute **official majority** (British officials) in the Central Legislative Council to ensure the government maintained control over all-India legislation.
- **However, it allowed a non-official majority in the Provincial Legislative Councils. (Note:** This "non-official" majority included both elected Indians and nominated non-officials, meaning the government still effectively held sway).
- **Additional Facts:**
  - ✓ **Communal Representation:** The most defining (and divisive) feature of the Act was the introduction of a system of communal representation for Muslims by accepting the concept of **'separate electorates'**. Lord Minto became known as the Father of Communal Electorate.
  - ✓ **Viceroy's Executive Council:** For the first time, it provided for the association of Indians with the executive councils of the Viceroy and Governors. **Satyendra Prasad Sinha** became the first Indian to join the Viceroy's Executive Council, appointed as the law member.
  - ✓ **Size Expansion:** It considerably increased the size of the legislative councils, both Central and provincial. The number of members in the Central Legislative Council was raised from 16 to 60.
  - ✓ **Deliberative Powers:** The Act enlarged the deliberative functions of the legislative councils at both levels. Members were now allowed to ask **supplementary questions** and move resolutions on the budget (though they still could not vote on the budget as a whole).

Q 93. Ans. B

Explanation:

- **Following the 1907 Surat Split, the British implemented a "carrot and stick" strategy:** pacifying the Moderates with the promise of constitutional reforms (the 1909 Act) while ruthlessly crushing the isolated Extremists.
- In 1908, **Bal Gangadhar Tilak**, the most prominent Extremist leader, was arrested for sedition over his writings in his newspaper Kesari. He was sentenced to six years in prison and deported to Mandalay (Burma), effectively leaving the Extremist movement leaderless and neutralized for years.
- **Additional Facts:**
  - ✓ **Fate of Other Leaders:** With Tilak in jail, the Extremist faction dissolved. **Aurobindo Ghosh** and **Bipin Chandra Pal** retired from active politics (Aurobindo moved to Pondicherry to

pursue spiritualism), and **Lala Lajpat Rai** left for abroad (Britain and later the USA).

- ✓ **Draconian Legislation:** To crush the Extremist press and public gatherings, the British enacted several harsh laws, including the **Seditious Meetings Act (1907)**, the **Indian Newspapers (Incitement to Offences) Act (1908)**, and the **Indian Press Act (1910)**.
- ✓ **The Alipore Bomb Case (1908):** Following an assassination attempt on a British judge by Khudiram Bose and Prafulla Chaki, the British arrested several nationalist leaders, including Aurobindo Ghosh, further breaking the Extremist network.
- ✓ **Tilak's Return:** The Extremist political vacuum lasted until Tilak was released in 1914. He re-entered Indian politics and launched the **Home Rule League** in 1916.

Q 94. Ans. B

Explanation:

- The early phase of the Indian National Congress (1885–1905) was dominated by the Moderates. Their primary political demand was the expansion and reform of the legislative councils to include more Indian representation.
- The British government conceded to some of these demands by passing the **Indian Councils Act of 1892**, making it the first constitutional achievement of the Moderate nationalists.
- **Additional Facts:**
  - ✓ **Indirect Elections:** The Act introduced the principle of representation for the first time, though the word "election" was carefully avoided. It used a system of "nomination on the recommendation" of bodies like universities, municipalities, and chambers of commerce.
  - ✓ **Budget Discussion:** For the first time, Indian members were granted the right to discuss the annual financial statement (the budget).
  - ✓ **Limitations:** While they could discuss the budget, members could **not** vote on it or move amendments. They were allowed to ask questions on public matters, but **supplementary questions were strictly prohibited**.
  - ✓ **Size Increase:** It expanded the number of additional (non-official) members in both the Central and Provincial Legislative Councils, though the official British majority was maintained.

Q 95. Ans. B

Explanation:

- Mahatma Gandhi officially launched the Civil Disobedience Movement by undertaking the historic **Dandi March** (Salt March) from the Sabarmati Ashram to the coastal village of Dandi. The movement began the moment he ceremonially broke the British salt monopoly on April 6, 1930.
- **The Lahore Session (1929)** is where the Congress authorized Gandhi to launch the movement and declared Purna Swaraj (Complete Independence), but the march itself was the physical beginning of the campaign.
- **Chauri Chaura (1922)** caused the end of the Non-Cooperation Movement.
- **First Round Table Conference (1930)** happened later and was boycotted by the Congress due to the ongoing Civil Disobedience Movement.
- **Additional Facts:**
  - ✓ **Route:** The march covered a distance of 240 miles (385 km) from Sabarmati Ashram in Ahmedabad to Dandi in Gujarat, lasting 24 days (March 12 – April 6, 1930).
  - ✓ **78 Followers:** Gandhi began the march with 78 chosen followers. By the time they reached Dandi, the crowd had swelled to tens of thousands.
  - ✓ **Why Salt?** Gandhi chose salt because it was universally consumed by both rich and poor. The oppressive British salt tax affected the poorest Indians the most, making it a powerful, unifying symbol of resistance.
  - ✓ **Parallel Salt Satyagrahas:**
    - **Tamil Nadu:** C. Rajagopalachari led a march from Tiruchirappalli to Vedaranyam.
    - **Malabar:** K. Kelappan led a march from Calicut to Payyanur.
    - **Dharasana:** Sarojini Naidu, Imam Saheb, and Manilal Gandhi led a massive non-violent raid on the Dharasana Salt Works (famous for the brutal police beatings witnessed by American journalist Webb Miller).
  - ✓ **Cunningham Circular:** In Assam, students played a major role in the movement by boycotting educational institutions in defiance of the Cunningham Circular, which forced students and guardians to provide guarantees of good behavior.

Q 96. Ans. C

Explanation:

- Announced by British Prime Minister Ramsay MacDonald in August 1932, the **Communal Award** not only continued the separate electorates already granted to Muslims, Sikhs, Indian Christians, Anglo-Indians, and Europeans, but it controversially extended them to the **Depressed Classes** (Dalits).
- Under a separate electorate, only voters from a specific community could vote for candidates of that community, which the British argued protected minority rights, but which Indian nationalists saw as a deliberate "divide and rule" tactic.
- **Additional Facts:**
  - ✓ **Gandhi's Fast:** Mahatma Gandhi strongly opposed the extension of separate electorates to the Depressed Classes, viewing it as a British attempt to permanently divide Hindu society. He began a "fast unto death" in the Yerwada Central Jail (Pune) in protest.
  - ✓ **Poona Pact (September 1932):** To save Gandhi's life, Dr. B.R. Ambedkar (who had advocated for separate electorates at the Round Table Conferences) negotiated an agreement with Hindu leaders. The Poona Pact abandoned separate electorates for the Depressed Classes in favor of **joint electorates with reserved seats**.
  - ✓ **Seat Increase:** The Poona Pact actually secured more political representation for the marginalized than the British award did. The number of reserved provincial seats for the Depressed Classes was increased from 71 (under the Communal Award) to 147.
  - ✓ **Central Legislature:** The Poona Pact also reserved 18% of the total seats for the Depressed Classes in the Central Legislature.

Q 97. Ans. A

Explanation:

- **Statement 1 is correct:** The Act continued and further expanded the British policy of communal representation (separate electorates) that was originally introduced in 1909 and expanded in 1919.
- **Statement 2 is correct:** Beyond just religious communities, the Act specifically extended the provision of separate electorates to new demographic groups, namely women and organized labour (workers), fracturing the electorate even further.
- **Statement 3 is incorrect:** The Act proposed an All-India Federation consisting of British Indian provinces and Princely States. However, this federation never actually materialized because the Princely States refused to join it.
- **Additional Facts:**
  - ✓ **Provincial Autonomy:** The Act abolished 'Dyarchy' (divided rule) in the provinces—which was introduced by the 1919 Act—and replaced it with Provincial Autonomy. Provinces were finally allowed to act as autonomous units of administration in their defined spheres.
  - ✓ **Dyarchy at the Centre:** While abolished in the provinces, the Act introduced Dyarchy at the Centre. Federal subjects were divided into 'reserved' (defense, external affairs) and 'transferred' subjects.
  - ✓ **Division of Powers:** It divided legislative powers into three lists: the Federal List, the Provincial List, and the Concurrent List.
  - ✓ **Residuary Powers:** Unlike modern India where residuary powers rest with the Parliament, the 1935 Act vested residuary powers directly in the hands of the **Viceroy**.
  - ✓ **Bicameralism:** It introduced bicameralism (two houses) in **6 out of 11 provinces**: Bengal, Bombay, Madras, Bihar, Assam, and the United Provinces.
  - ✓ **New Institutions:** The Act provided for the establishment of the **Reserve Bank of India** (to control currency and credit), the **Federal Court** (which was set up in 1937 in Delhi), and Provincial/Joint Public Service Commissions.

Q 98. Ans. C

Explanation:

- **Statement 1 is correct:** The CR Formula explicitly stated that the terms regarding demarcation and separation would only become operative after the end of the Second World War and Britain's complete transfer of power to India. In the interim, the Muslim League was expected to endorse the Congress demand for independence and cooperate in forming a provisional government.
- **Statement 2 is correct:** The formula proposed that after the war, a commission would demarcate contiguous districts in the Northwest and Northeast of India where Muslims were in an absolute majority. A plebiscite (voting) of all adult inhabitants (not just Muslims) in these areas would then decide the issue of separation.

- **Additional Facts:**
  - ✓ **The Objective:** Formulated by C. Rajagopalachari in 1944, it was a tacit acceptance of the concept of Pakistan and an attempt to break the political deadlock between the Indian National Congress and the All-India Muslim League.
  - ✓ **Gandhi-Jinnah Talks (1944):** Mahatma Gandhi backed the CR Formula and used it as the basis for his negotiations with Muhammad Ali Jinnah. The talks ultimately failed.
  - ✓ **Why Jinnah Rejected It:** Jinnah rejected the formula for several reasons:
    - He wanted the plebiscite to be restricted **only to the Muslim population** of those regions, not all adult inhabitants.
    - He demanded a complete partition before the British left, rather than a transition under a joint interim government.
    - He opposed the formula's suggestion of mutual agreements on essential matters like defense, commerce, and communications in the event of separation.
  - ✓ **Other Opposition:** The formula was also vehemently opposed by the Hindu Mahasabha and leaders like V.D. Savarkar, who were against any form of partition of the Indian subcontinent.

Q 99. Ans. A

Explanation:

- **R.C. Majumdar:** He made this famous concluding remark in his book The Sepoy Mutiny and the Revolt of 1857. He argued it was not "first" due to previous mutinies (like the Vellore Mutiny of 1806), not "national" because large parts of India and many societal classes did not participate, and not a "war of independence" because the rebels lacked a unified vision of a free India and fought primarily for local grievances or lost privileges.
- **V.D. Savarkar:** In his 1909 book The Indian War of Independence 1857, he was the first to interpret the revolt not as a mere mutiny, but as a planned and organized political and national struggle for freedom.
- **S.N. Sen:** Commissioned by the Government of India to write the official history of the revolt (Eighteen Fifty-Seven), Sen concluded that what began as a fight for religion ended as a war of independence, though he stopped short of calling it a fully unified "national" war.
- **Tara Chand:** He characterised the revolt as a transitionary event, the last attempt of the old conservative order (feudal lords and princes) to regain their lost power, rather than a modern nationalistic movement.
- **Additional Facts:**
  - ✓ **Benjamin Disraeli:** The contemporary British Conservative leader in Parliament famously described the uprising as a "national revolt" rooted in deep systemic grievances, rejecting the official British line that it was merely a sepoy mutiny.
  - ✓ **John Lawrence & John Seeley:** British historians who completely dismissed the political and nationalist angles, calling it nothing more than a "wholly unpatriotic and selfish Sepoy Mutiny" with no native leadership and no popular support.
  - ✓ **Karl Marx:** Writing for the New York Tribune as the events were unfolding, Marx recognized the uprising not just as a military mutiny, but as a broader "national revolt" against British colonial exploitation.
  - ✓ **T.R. Holmes:** Represented the racist imperial viewpoint by describing the revolt as a conflict between "civilization and barbarism."

Q 100. Ans. D

Explanation:

- **Keshab Chandra Sen** was the prominent social reformer who famously stated, "Whoever worships the True God daily must learn to recognise all his fellow countrymen as brethren".
- This quote reflects the universalist, egalitarian, and humanistic philosophy that he sought to infuse into the Brahmo movement, emphasizing that true religious devotion must necessarily translate into social equality, the eradication of caste barriers, and universal brotherhood.
- **Additional Facts:**
  - ✓ **The First Split (1866):** Sen joined the Brahmo Samaj in 1857. However, his radical views, such as completely discarding the sacred thread, promoting inter-caste marriages, and advocating for women's active participation, led to a clash with the more conservative Debendranath Tagore. This resulted in Sen forming the **Brahmo Samaj of India**, while Tagore's faction became the **Adi Brahmo Samaj**.
  - ✓ **Indian Reform Association (1870):** Founded by Sen to create public opinion against child

marriage, elevate the social status of women, and promote secular education.

- ✓ **Native Marriage Act (1872):** Also known as the Brahmo Marriage Act, Sen was instrumental in pushing the British government to pass this legislation. It legalized inter-caste marriages and set a minimum marriageable age (14 for girls, 18 for boys).
- ✓ **The Second Split (1878):** Ironically, a second split occurred when Sen married his own 13-year-old daughter to the minor Hindu Maharaja of Cooch Behar with traditional Vedic rites, violating his own rules against child marriage and idolatry. His disillusioned followers broke away to form the **Sadharan Brahmo Samaj** (led by Ananda Mohan Bose and Shibnath Shastri).
- ✓ **Nababidhan (1881):** Following the second split, Sen established the Nababidhan (New Dispensation), a new religious movement attempting to synthesize the core tenets of Hinduism, Christianity, Buddhism, and Islam into one universal religion.

Q 101. Ans. D

Explanation:

- **Pair 1 is correct:** Sandhya was a fiery revolutionary newspaper published in **Bengal**, founded by Brahmabandhab Upadhyay during the height of the Swadeshi movement (1906).
- **Pair 2 is correct:** The Indian Sociologist was an early nationalist journal published in **London**, founded by Shyamji Krishna Varma in 1905 to promote the cause of Indian self-rule and support the activities of India House.
- **Pair 3 is correct:** Talvar (officially Madan's Talvar, named in honor of the martyred Madan Lal Dhingra) was published from **Berlin** by Virendranath Chattopadhyaya (with support from Madame Bhikaji Cama) around 1909-1910.
- **Pair 4 is correct:** Free Hindustan was a pioneering anti-British journal published initially from **Vancouver** (Canada) and later from Seattle (USA) by Tarak Nath Das in 1908. It was the first truly revolutionary Indian journal in North America.
- **Additional Facts:**
  - ✓ **Ghadar:** The most famous revolutionary newspaper published abroad, starting in 1913 from San Francisco (USA) by the Ghadar Party. Its inaugural issue was in Urdu, followed by Gurmukhi, and it prominently carried the subtitle Angrezi Raj Ka Dushman (An Enemy of the British Rule).
  - ✓ **Bande Mataram (Paris):** An Indian nationalist publication launched in Paris in 1909 by **Madame Bhikaji Cama**. (Note: This is distinct from the Bande Mataram English daily published in India by Bipin Chandra Pal and Aurobindo Ghosh).
  - ✓ **Yugantar (Jugantar):** A highly influential Bengali revolutionary weekly started in 1906 by Barindra Kumar Ghosh, Abhinash Bhattacharya, and Bhupendranath Datta (Swami Vivekananda's brother). It became the ideological mouthpiece of the Anushilan Samiti.
  - ✓ **Kal:** A radical Marathi weekly published in Pune (Maharashtra) by Shivram Mahadev Paranjape, which fiercely supported the Extremist faction and revolutionary activities.

Q 102. Ans. A

Explanation:

- **Moplah Uprisings (1836–1854):** The Moplah (Muslim) tenants of the Malabar region (Kerala) waged a series of violent uprisings against the oppressive Hindu landlords (jenmis) and British revenue officials. (Note: While the most famous Khilafat-linked Moplah Rebellion occurred in 1921, the initial and continuous series of uprisings began much earlier, in 1836).
- **Indigo Revolt (1859–1860):** Also known as the Nil Bidroha, this occurred in Bengal. Peasants, led by Digambar Biswas and Bishnu Biswas, refused to grow indigo under the exploitative terms forced by European planters.
- **Deccan Peasants' Uprising (1875):** Agrarian riots broke out in the Poona and Ahmednagar districts of Maharashtra. The Maratha peasants rose against corrupt Marwari and Gujarati moneylenders who were seizing their lands through debt traps.
- **Phadke's Ramosi Uprising (1879):** Vasudev Balwant Phadke gathered the Ramosi peasant community in Maharashtra and organized an armed rebellion. He fought against the British failure to take up anti-famine measures and aimed to establish an independent Indian republic.
- **Additional Facts:**
  - ✓ **Indigo Commission:** The success of the Indigo Revolt was partly due to the support of the Bengali intelligentsia (like Harish Chandra Mukherjee). It forced the British to appoint the Indigo Commission in 1860, which ruled in favor of the peasants.

- ✓ **Deccan Agriculturists' Relief Act (1879):** The Deccan riots were largely successful. The government appointed a commission to investigate the peasants' grievances, resulting in this 1879 act that placed restrictions on the alienation of peasant land.
- ✓ **Pabna Agrarian League (1873):** During the same period as the Deccan riots, peasants in the Pabna district of East Bengal formed an agrarian league to resist the oppressive extraction of abwabs (illegal cesses) by zamindars.
- ✓ **Bankim Chandra Chatterjee:** His famous novel Anandamath was based on the earlier **Sanyasi Rebellion** (1763–1800), which occurred in Bengal after the catastrophic famine of 1770.

Q 103. Ans. B

Explanation:

- **Gulbadan Begum**, the daughter of Emperor Babur and the half-sister of Emperor Humayun, wrote the Humayun-nama.
- She wrote this biographical account at the request of her nephew, Emperor Akbar, who wanted to collect source material for his own official history (the Akbarnama). Her work provides a unique and intimate look into the domestic life of the early Mughal royal household, rather than just focusing on military and political conquests.
- **Additional Facts:**
  - ✓ **Abul Fazl:** The grand vizier of Emperor Akbar and the author of the **Akbarnama** (the official history of Akbar's reign in three volumes). The third and most famous volume is the **Ain-i-Akbari**, which details the administration, army, and revenue system of Akbar's empire.
  - ✓ **Abdul Qadir Badauni:** A historian and translator at Akbar's court, but also a staunch orthodox critic of Akbar's liberal religious policies (like the Din-i-Ilahi). He wrote the **Muntakhab-ut-Tawarikh**, a general history of India that was kept secret until Akbar's death due to its highly critical nature.
  - ✓ **Baburnama (Tuzuk-i-Baburi):** The autobiography of the founder of the Mughal Empire, Zahir-ud-Din Muhammad Babur. Unlike most later Mughal texts which were in Persian, Babur wrote this in his mother tongue, **Chagatai Turkic**.
  - ✓ **Tuzuk-i-Jahangiri:** The fascinating and remarkably candid Persian autobiography of Emperor Jahangir, detailing his reign, his fascination with art and nature, and his struggles with rebellion and addiction.

Q 104. Ans. C

Explanation:

- Following the abrupt withdrawal of the Non-Cooperation Movement and the collapse of the Khilafat Movement in 1922, the temporary Hindu-Muslim unity of the early 1920s shattered. The period leading up to the Civil Disobedience Movement (1930) was marked by a severe wave of communal violence across British India.
- **Multan (1922):** Major riots broke out during the Muharram festival.
- **Amritsar (1923):** Witnessed severe communal clashes alongside places like Saharanpur and Sindh.
- **Delhi & Aligarh (1924):** The year 1924 was particularly bloody, with devastating communal riots occurring in Delhi, Aligarh, Lucknow, Nagpur, and the infamous Kohat riots.
- Therefore, all four listed places experienced significant communal riots during the decade preceding the Civil Disobedience Movement.
- **Additional Facts:**
  - ✓ **Gandhi's 21-Day Fast:** In September 1924, deeply distressed by the brutal communal riots in Kohat (North-West Frontier Province), Mahatma Gandhi undertook a 21-day fast in Delhi to restore Hindu-Muslim harmony.
  - ✓ **Shuddhi and Tabligh Movements:** The 1920s saw aggressive religious mobilization. The Arya Samaj launched the Shuddhi movement (to reconvert those who had converted to Islam) and the Sangathan movement (to organize Hindus). In response, Muslim leaders launched the Tabligh (propaganda) and Tanzim (organization) movements, further polarizing society.
  - ✓ **Rise of Communal Organizations:** This volatile decade witnessed the revitalization of the **Hindu Mahasabha** (which held its major session in 1923 under Madan Mohan Malaviya) and the founding of the **Rashtriya Swayamsevak Sangh (RSS)** in 1925 by K.B. Hedgewar. On the other side, the **Muslim League** became more uncompromising in its demands for separate political rights.
  - ✓ **Simon Commission Boycott (1927):** Interestingly, it was the all-white Simon Commission that temporarily forced a fragile political unity among Indians again, as both

Congress and a faction of the Muslim League (led by Jinnah) boycotted it, paving the way for the Nehru Report (1928).

Q 105. Ans. A

Explanation:

- **Satish Chandra Samanta** was the first Sarbadhinayak (Supreme Leader) of the **Tamluk Jatiya Sarkar** (National Government).
- Established in **December 1942** in the **Medinipur** district of Bengal during the Quit India Movement, this parallel government was one of the most successful and longest-lasting independent local governments set up by Indian nationalists, functioning until **August 1944.**
- Following Samanta's arrest in 1943, the leadership was passed to others, including **Ajoy Mukherjee**.
- **Additional Facts:**
  - ✓ **Vidyut Vahini:** The Tamluk Jatiya Sarkar maintained its own armed militia and police force known as the Vidyut Vahini (Lightning Fleet), which was commanded by **Sushil Kumar Dhara**. It also ran its own postal system, courts, and famine-relief programs.
  - ✓ **Matangini Hazra:** A 73-year-old Gandhian affectionately known as Gandhi Buri. Before the Jatiya Sarkar was fully established, she was shot dead by British police in September 1942 while bravely leading a procession holding the Indian tricolor in Tamluk.
  - ✓ **Ballia (Uttar Pradesh):** The first parallel government of the Quit India Movement was established here under the leadership of **Chittu Pandey**. However, it was short-lived, lasting only a few weeks.
  - ✓ **Satara (Maharashtra):** The most famous and **longest-lasting** parallel government, known as the Prati Sarkar. Led by **Nana Patil** (and supported by leaders like Y.B. Chavan), it ran effectively from mid-1943 to 1945. It had volunteer militias (Seva Dals) and established 'Nyayadan Mandals' (people's courts).
  - ✓ **Talcher (Odisha):** Another notable parallel government was formed here under the leadership of Laxman Nayak.

Q 106. Ans. A

Explanation:

- **Statement 1 is correct:** In the 1937 provincial elections (held under the provisions of the Government of India Act, 1935), the Indian National Congress swept the polls. While they secured an absolute majority in five provinces (Madras, United Provinces, Central Provinces, Bihar, and Orissa), they emerged as the **single largest party** in four others: Bombay, Assam, Bengal, and the North-West Frontier Province (NWFP).
- **Statement 2 is incorrect:** Congress performed poorly in Punjab and Sindh. In Punjab, the **Unionist Party** won a sweeping majority. In Sindh, regional Muslim coalitions (such as the Sind United Party) dominated the elections.
- **Additional Facts:**
  - ✓ **Total Provinces:** Elections were held in 11 provinces. Congress eventually formed governments in **8 out of 11 provinces** (adding Bombay, Assam, and NWFP to their five majority provinces).
  - ✓ **Non-Congress Governments:**
    - **Bengal:** A coalition government was formed by the **Krishak Praja Party** (led by A.K. Fazlul Huq) and the All-India Muslim League.
    - **Punjab:** The government was formed by the **Unionist Party** (led by Sikandar Hayat Khan).
    - **Sindh:** A fractured mandate led to a coalition government under Sir Ghulam Hussain Hidayatullah (and later Allah Bux Soomro).
  - ✓ **Resignation of Ministries (1939):** The Congress ministries governed for about 28 months. They all abruptly resigned in October–November 1939. This was in protest against Viceroy Lord Linlithgow's unilateral decision to drag India into World War II without consulting Indian political leaders.
  - ✓ **Day of Deliverance:** Muhammad Ali Jinnah and the Muslim League celebrated the mass resignation of the Congress ministries as the 'Day of Deliverance' (Yaum-e-Nijat) on December 22, 1939, receiving support from Dr. B.R. Ambedkar and the Justice Party.

Q 107. Ans. B

Explanation:

- The **Defence of India Act, 1939** (and the Rules promulgated under it) was an emergency colonial legislation enacted immediately after the outbreak of the Second World War.
- Its primary objective was to grant the British colonial government absolute and sweeping

powers to ensure the unhindered extraction of Indian resources for the war effort. It allowed the government to bypass regular legal procedures to suppress any nationalist political dissent, censor the press, and indefinitely detain political leaders without trial.

- **Additional Facts:**
  - ✓ **World War I Predecessor:** It was heavily modeled on the earlier **Defence of India Act, 1915**, which was similarly enacted during the First World War to crush revolutionary activities (like the Ghadar movement) and later formalized into the draconian Rowlatt Act of 1919.
  - ✓ **Quit India Movement (1942):** The Defence of India Rules were the primary legal instrument used by the British to ruthlessly crush the Quit India Movement. Under these rules, the entire top brass of the Indian National Congress (including Mahatma Gandhi, Jawaharlal Nehru, and Sardar Patel) was arrested overnight in "Operation Zero Hour" on August 9, 1942.
  - ✓ **Economic Impact:** The rules gave the government total control over the Indian economy, allowing them to fix prices, commandeer transport, and divert essential goods (like foodgrains) for military use. This diversion of resources significantly contributed to the catastrophic **Bengal Famine of 1943**.
  - ✓ **Post-War Transition:** Even after the war ended in 1945, the continuation of these oppressive wartime laws fueled intense public anger, which eventually erupted during the INA Trials and the Royal Indian Navy (RIN) Mutiny of 1946. Some of its preventive detention provisions controversially found their way into the legal framework of independent India.

Q 108. Ans. A

Explanation:

- **Statement 1 is correct:** The first session of the Indian National Congress (INC) was attended by exactly 72 delegates representing different provinces of British India.
- **Statement 2 is incorrect:** The first session was presided over by **Womesh Chandra Bonnerjee** (W.C. Bonnerjee), a prominent Bengali lawyer. Dadabhai Naoroji presided over the second session of the INC, which was held in Calcutta in 1886 (and he later presided in 1893 and 1906).
- **Additional Facts:**
  - ✓ **Venue Change:** The inaugural session was originally scheduled to be held in Poona (Pune). However, due to an outbreak of cholera in the city, the venue was abruptly shifted to the **Gokuldas Tejpal Sanskrit College** in Bombay, held from December 28–30, 1885.
  - ✓ **O. Hume & The Safety Valve Theory:** The INC was founded by Allan Octavian Hume, a retired British civil servant. Extremist leaders like Lala Lajpat Rai later propagated the "Safety Valve Theory," arguing that Hume and Viceroy Lord Dufferin created the Congress to provide a safe, peaceful outlet for growing Indian discontent and prevent another revolt like 1857.
  - ✓ **Lord Dufferin's Stance: Lord Dufferin** was the Viceroy of India when the INC was founded. While initially tolerant, he soon became hostile to the organization, famously mocking it in 1887 as representing only a "microscopic minority" of the Indian population.
  - ✓ **First Woman Graduate:** In 1890 (the sixth session), Kadambini Ganguly, the first female graduate of Calcutta University, addressed the Congress session, marking a significant milestone for women's participation in the national movement.

Q 109. Ans. D

Explanation:

- **Trial of Bahadur Shah Zafar (1858):** Following the suppression of the 1857 Revolt, the last Mughal Emperor was tried for treason and rebellion by a British military commission. He was found guilty and exiled to Rangoon (Yangon), Myanmar.
- **Trial of Surendranath Banerjea (1883):** Banerjea became the first Indian journalist to be imprisoned. He was tried for contempt of court after writing an angry editorial in his newspaper, The Bengalee, criticizing a British judge (Justice Norris) for bringing a Hindu idol into the courtroom.
- **Trial of Bal Gangadhar Tilak (1897 & 1908):** Tilak faced multiple famous sedition trials. The first major one was in 1897 for his writings in Kesari defending Shivaji's killing of Afzal Khan. He was tried again in 1908 and deported to Mandalay for six years.
- **Trial of V.D. Savarkar (1910–1911):** Arrested in London in 1910 and extradited to India (famously

attempting escape at Marseilles), Savarkar was tried in the Nasik Conspiracy Case for abetting the assassination of A.M.T. Jackson. He was sentenced to two life terms (50 years) in the Cellular Jail (Kalapani).

- **Additional Facts:**
  - ✓ **Section 124A (Sedition):** Inserted into the Indian Penal Code in 1870, this became the primary legal weapon the British used against Indian nationalists. Tilak, Gandhi, and Annie Besant were all famously tried under it.
  - ✓ **The Alipore Bomb Trial (1908–1909):** One of the most famous early revolutionary trials following the Muzzafarpur bombing attempt by Khudiram Bose and Prafulla Chaki. Aurobindo Ghosh was a prime accused but was successfully defended by **Chittaranjan Das** and acquitted.
  - ✓ **The Meerut Conspiracy Case (1929–1933):** A massive trial targeting the rising Communist and trade union movement in India. The British arrested 31 prominent labor leaders (including three Englishmen), drawing international condemnation and sympathy for the communist cause.
  - ✓ **INA Trials (1945–1946):** Held at the Red Fort, Delhi, the British tried officers of Subhas Chandra Bose's Indian National Army (Prem Kumar Sahgal, Shah Nawaz Khan, and Gurbaksh Singh Dhillon) for treason. The trials sparked massive nationalist uprisings across India, forcing the British to eventually remit their sentences.

Q 110. Ans. D

Explanation:

- **Statement 1 is incorrect:** The Hunter Education Commission was appointed to review the progress of education in the country since Wood's Despatch of 1854. However, its purview was strictly confined to **primary and secondary education**. University education was largely excluded from its mandate (the major evaluation of universities came later with the Raleigh Commission of 1902).
- **Statement 2 is correct:** A landmark recommendation of the Commission was that the control, management, and funding of primary education should be transferred to the newly formed local bodies, namely the District and Municipal Boards. This aligned perfectly with the Viceroy's push for local self-government.
- **Additional Facts:**
  - ✓ **The Viceroy:** The Commission was appointed in 1882 by **Lord Ripon**, who is also known as the "Father of Local Self Government in India." It was chaired by Sir William Wilson Hunter.
  - ✓ **Bifurcation of Secondary Education:** The Commission recommended dividing secondary education into two distinct streams: a **literary stream** (leading to the university entrance examination) and a **vocational stream** (preparing students for commercial or practical careers).
  - ✓ **Role of Private Enterprise:** It strongly advocated that the government should gradually withdraw from direct management of higher education institutions. Instead, it should encourage private enterprise to step in, supported by a robust system of government grants-in-aid.
  - ✓ Hunter Education Commission (1882) was distinct from the **Hunter Committee of Inquiry (1919)**. The latter was the Disorders Inquiry Committee formed to investigate the Jallianwala Bagh massacre in Punjab.

Q 111. Ans. B

Explanation:

- The national-level celebration for World Elephant Day was hosted in **Coimbatore**, Tamil Nadu.
- The event was organized by the Ministry of Environment, Forest and Climate Change (MoEFCC) to highlight India's role as the custodian of nearly 60% of the global wild Asian elephant population.
- During the celebration, the prestigious **Gaj Gaurav Awards** were presented to recognize the exemplary, grassroots-level contributions of frontline forest staff, mahouts, and local communities in protecting elephants and mitigating human-elephant conflict.
- **Additional Facts:**
  - ✓ **World Elephant Day:** Celebrated globally on **August 12** every year to spread awareness about the preservation and protection of elephants.
  - ✓ **Project Elephant:** Launched by the Government of India in **1992** as a Centrally Sponsored Scheme to protect elephants, their habitats, and corridors, and to address issues of human-animal conflict.
  - ✓ **National Heritage Animal:** The elephant (Elephas maximus) was declared India's National Heritage Animal in 2010 to scale up its protection status.

- ✓ **Recent Venues:** For context on other recent years, the national-level celebrations were held in Bhubaneswar, Odisha (2023) and the Periyar Tiger Reserve, Kerala (2022).
- ✓ **Gaj Utsav:** A separate major conservation festival, Gaj Utsav, was inaugurated by the President of India at Kaziranga National Park (Assam) in 2023 to commemorate 30 years of Project Elephant.

Q 112. Ans. B

Explanation:

- India's first wildlife-safe highway stretch featuring innovative "table-top red markings" was introduced on **National Highway 45 (NH-45)** in Madhya Pradesh.
- This 11.96-kilometre stretch (the Hiran-Sindoor section between Bhopal and Jabalpur) passes through a dense and ecologically sensitive forest corridor flanked by the Nauradehi Wildlife Sanctuary and the Veerangana Durgavati Tiger Reserve.
- The 5mm thick, textured red markings act as a visual and tactile warning, naturally encouraging drivers to slow down without sudden braking, thereby significantly reducing the risk of animal-vehicle collisions in the wildlife corridor.
- **Additional Facts:**
  - ✓ **Design Inspiration:** The concept of the table-top red markings was inspired by road safety designs used internationally, particularly on Sheikh Zayed Road in Dubai.
  - ✓ **Other Wildlife Features:** Along with the red markings, the NHAI has constructed approximately 25 dedicated animal underpasses and erected 8-foot-high fencing on this stretch to allow tigers, deer, and other animals to cross the highway safely without entering the carriageway.
  - ✓ **Green Highways Policy:** This initiative falls under the Ministry of Road Transport and Highways' (MoRTH) Green Highways (Plantation, Transplantation, Beautification & Maintenance) Policy launched in 2015, which aims to balance infrastructure growth with ecological conservation.
  - ✓ **India's Largest Elevated Wildlife Corridor:** While the NH-45 project introduced red markings, the **Delhi-Dehradun Expressway** features Asia's largest elevated wildlife corridor (12 km long) passing over the Rajaji National Park, designed to allow unhindered movement of elephants and other wildlife underneath.

Q 113. Ans. B

Explanation:

The "No Kings" protest movement is a series of large-scale demonstrations that took place primarily across the United States, opposing the actions and policies of the second Trump administration. The movement's name reflects its core message, that the U.S. president does not hold monarchical or absolute power, and that "America has No Kings" and "the power belongs to the people."

Q 114. Ans. B

Explanation:

- **PRISM-SG Portal — full form:** Portal for Rail-Road Inspection & Stages Management (Steel Girders) — was jointly launched on 25 March 2026 in New Delhi by Union Minister Nitin Gadkari (Road Transport and Highways) and Union Minister Ashwini Vaishnaw (Railways).
- **Purpose:** It digitises the end-to-end approval and inspection process for Road Over Bridges (ROBs) and railway bridges, covering steps like Quality Assurance Plan (QAP) submission, Welding Procedure Specification Sheets (WPSS), and fabrication-stage inspections of steel girders, processes that were earlier handled offline and outside the existing RRCAS (Rail-Road Crossing Approval System) portal.
- **Stakeholders integrated:** Road Owning Departments, Indian Railways (including Zonal Railways and RDSO), contractors, fabricators, and inspection agencies, all brought onto a single digital platform with a complete audit trail.

Q 115. Ans. D

Explanation:

- **Pair 1 is correct:** Kami Rita Sherpa scaled Everest for a record-breaking 32nd time on 17 May 2026, extending his own world record for the most Everest ascents, reaching the summit at 10:12 am while leading a 14 Peaks Expedition team.
- **Pair 2 is correct:** Lionel Messi officially became a billionaire in May 2026 per the Bloomberg Billionaires Index, becoming the second footballer in history to reach this milestone, after Cristiano Ronaldo (who became the first following his 2023 move to Al-Nassr).
- **Pair 3 is correct:** Mitchell Starc was named Wisden's Leading Men's Cricketer in the World and Deepti Sharma the Leading Women's Player, announced

ahead of the 163rd Wisden Cricketers' Almanack (2026 edition) — Starc for his outstanding 2025 (55 Test wickets, key Ashes role), and Deepti Sharma for her Player-of-the-Tournament showing in India's maiden ODI Women's World Cup win.

Q 116. Ans. A

Explanation:

- **FIDE Women's Candidates Tournament winner → 2 (R. Vaishali):** GM Vaishali Rameshbabu won the 2026 FIDE Women's Candidates in Cyprus, defeating Kateryna Lagno in the final round to finish outright first on 8.5/14, earning the right to challenge reigning champion Ju Wenjun for the Women's World Championship.
- **India's 95th Grandmaster → 1 (Aronyak Ghosh):** The Kolkata-based player secured his third and final GM norm at the Bangkok Chess Club Open 2026, becoming India's 95th Grandmaster and West Bengal's 12th.
- **Second-youngest Grandmaster in history → 4 (Faustino Oro):** The Argentine "Messi of Chess" clinched his third GM norm at the Sardinia International Chess Festival (May 2026) at age 12 years, 6 months, 26 days — the second-youngest ever, behind only Abhimanyu Mishra (12 years, 4 months, 25 days) and ahead of Sergey Karjakin.
- **Secured International Master title → 3 (Apar Saxena):** Emerging Indian chess star Apar Saxena successfully earned his International Master (IM) title in 2026.By elimination, and consistent with recent Indian chess news cycles of young players earning IM titles.
- **Additional Facts:**
  - ✓ **Faustino Oro's Records:** Often dubbed the "Messi of Chess," Oro previously made history by becoming the youngest-ever player to achieve the International Master title at age 10. The only player to reach the Grandmaster title faster than Oro is Abhimanyu Mishra (12 years, 4 months, 25 days).
  - ✓ **India's GM Boom:** Aronyak Ghosh's achievement highlights the rapid growth of chess in India, becoming the 12th Grandmaster from the state of West Bengal alone. The player who started this legacy, Viswanathan Anand, became India's 1st GM in 1988.

Q 117. Ans. C

Explanation:

- **Gurindervir Singh → 2 (National record in 100 metres, 10.09 seconds):** The Punjab-born Indian Navy sprinter clocked 10.09 seconds at the Federation Cup in Ranchi (23 May 2026), becoming the first Indian to breach the 10.10-second barrier in the 100m — breaking the national record twice within 24 hours.
- **Vishal T.K. → 3 (Broke the 45-second barrier in the 400m, 44.98 seconds):** At the same Federation Cup meet, Vishal became the first Indian to complete the 400m in under 45 seconds, recording Asia's fastest 400m time of the season.
- **Ishank Singh → 1 (Swam the Palk Strait in 9 hours 50 minutes):** The seven-year-old from Ranchi swam the 29 km stretch from Talaimannar (Sri Lanka) to Dhanushkodi (India) on 30 April 2026, becoming the youngest and fastest person ever to complete the crossing, officially recognised by the Universal Records Forum.
- **Sebastian Sawe → 4 (Completed a marathon in under two hours):** The Kenyan runner made history at the London Marathon on 26 April 2026, winning in 1:59:30, becoming the first person ever to run an official, record-eligible marathon under two hours (unlike Eliud Kipchoge's famous 1:59:40 in 2019, which was run in a specially arranged, non-record-eligible event with rotating pacemakers).

Q 118. Ans. C

Explanation:

- The Indian footballer who won the All India Football Federation (AIFF) Men's Player of the Year Award for the 2024-25 season is **Subhasish Bose.**
- Subhasish Bose, a defender and captain for Mohun Bagan Super Giant, received this honor at the AIFF Awards ceremony held in Bhubaneswar on May 2, 2025. He was recognized for his pivotal role in leading his club to both the Indian Super League (ISL) League Shield and the ISL Cup during a record-breaking season.
- **Brison Fernandes (FC Goa)** was named the AIFF "Promising Men's Player of the Year" for the 2024-25 season.
- **Lallianzuala Chhangte** and **Sunil Chhetri** are previous winners of the Men's Player of the Year award, with Chhetri holding the record for the most wins (seven).

Q 119. Ans. B

Explanation:

- **Operation Himsetu (Trishakti Corps) → Sikkim** — launched by the Trishakti Corps (Eastern

Command) in April 2026 to evacuate tourists stranded near Lachen-Chungthang, North Sikkim, after a landslide.

- **Operation Netra 1.0 → Ladakh** — a four-day advanced surgical eye camp held 27–30 April 2026 at the 153 General Hospital, Leh, Ladakh.
- **Brahmastra live-firing exercise** → Jaisalmer sector, Rajasthan — an Apache AH-64E firepower demonstration held at the Pokhran Firing Range, Jaisalmer district, in April 2026.
- **Pragati-2026 multinational exercise** → Umroi, Meghalaya — the maiden edition of this 12/13-nation exercise, held at Umroi Military Station, Meghalaya, from 20 May 2026.

Q 120. Ans. B

Explanation:

Malaysia launched Ryt Bank on 25 August 2025, just ahead of Merdeka (Independence Day), billed as the world's first fully licensed, AI-powered digital bank. It was jointly launched by YTL Group (via YTL Power International) in partnership with Singapore's Sea Limited, and was the last of five digital banks licensed by Bank Negara Malaysia (BNM) to go live, following GX Bank, Boost Bank, AEON Bank, and KAF Digital Bank.

Q 121. Ans. D

Explanation:

- The Employees' Provident Fund Organisation (EPFO), under the Ministry of Labour & Employment, increased the auto-settlement limit for advance PF claims from ₹1 lakh to ₹5 lakh, as part of its broader "EPFO 3.0" digital transformation initiative.
- **Key details:**
  - ✓ The proposal to enhance the limit was approved at a meeting held in Srinagar in March 2025, and subsequently cleared by the Central Board of Trustees.
  - ✓ This is part of a progressive increase — the auto-settlement facility began with just illness/hospital advances, was later extended to marriage, education, and housing claims, and the limit itself rose from ₹50,000 3 ₹1 lakh 3 and now ₹5 lakh.
  - ✓ With this raised limit, EPFO has been auto-settling around 70% of all advance claims without manual scrutiny, and such eligible claims are now processed within 3 days, backed by a system of 12% penal interest on officials responsible for unjustified delays beyond 20 days.
  - ✓ This reform is part of a larger EPFO overhaul under the new Employees' Provident Funds Scheme, 2026 (along with the Employees' Pension Scheme, 2026, and EDLI Scheme, 2026), which also reduced advance withdrawal categories from 13 to just three (illness, education, marriage) and cut minimum service requirements for several withdrawal types.

Q 122. Ans. D

Explanation:

- The Patharkatti Stone Craft (also spelled Pattharkatti) originates from Patharkatti village, located in the Neemchak Bathani block of Gaya district, Bihar. It received the GI tag on 20 February 2025.
- **Key facts:**
  - ✓ The craft is nearly 300 years old, and involves intricate carvings on locally-sourced black granite, depicting Lord Buddha, Goddess Mahagauri, various deities, and folk-culture figures.
  - ✓ According to local tradition, Maratha ruler Ahilyabai Holkar brought around 900 Gour Brahmin artisans from Rajasthan to the region to help reconstruct the Vishnupad Temple at Gaya (completed in 1787), and it's believed the village itself was named "Pattharkatti" (meaning "stone-cutting") in connection with this history. The Vishnupad Temple is said to have been built using black granite sourced from this very village.
  - ✓ Over 650 artisans in Patharkatti are currently associated with this stone-carving tradition.
  - ✓ **This GI tag was granted alongside two other Bihar crafts in the same batch:** Bawan Buti sarees/fabric (Nalanda district) and Pidiya/Peedhiya painting (Bhojpur district), achieved through the joint efforts of NABARD and the Bihar Government.

Q 123. Ans. D

Explanation:

- **Statement 1 is correct:** IIT Kharagpur did inaugurate the Vikram Sodhi Centre of Excellence for AI-Enabled Geological and Mining Systems on 17 April 2026, an interdisciplinary research initiative backed by a ₹15 crore, five-year funding commitment from Vikram Sodhi, Vice Chairman of Mineros SA and Managing Partner at Sun Valley Investments. The centre focuses on applying AI

across the full mining value chain, exploration, mine planning, processing, predictive maintenance, and ESG analytics, tailored to Indian geological conditions.

- **Statement 2 is correct:** India's first quantum computer test beds, named Amaravati 1S and Amaravati 1Q, were inaugurated by Andhra Pradesh CM N. Chandrababu Naidu on 14 April 2026 at SRM University, Amaravati, as part of the state's "Quantum Valley" initiative. Officials described this as India's first indigenous, open-access quantum computer facility. (Building on this, IBM has since also confirmed it will deploy a full-scale physical quantum computer at the same Amaravati Quantum Valley hub by September 2026, reinforcing Amaravati's positioning as India's "Quantum Capital.")

Q 124. Ans. A

Explanation:

- **Statement 1 is correct:** Microsoft has indeed introduced MAI-Transcribe-1, an in-house speech-to-text AI model built by the Microsoft AI Superintelligence team, supporting 25 languages and powering features like Copilot's Voice Mode and dictation. It was introduced in 2025-26 and is available via Microsoft Foundry/Azure Speech.
- **Statement 2 is incorrect:** IIT Guwahati researchers did develop a new predictive framework (using a Bayesian Neural Network combined with high-resolution satellite imagery and digital elevation models) to monitor glacier hazards and glacial lake formation, but it was developed for the Eastern Himalayas, not the Western Himalayas. The study, led by Prof. Ajay Dashora (Department of Civil Engineering), identified 492 locations in the Eastern Himalaya where new glacial lakes are likely to form, aiding early-warning systems for Glacial Lake Outburst Floods (GLOFs).

Q 125. Ans. D

Explanation:

- **Pair 1 is correctly matched:** Rithika Sri from Salem, Tamil Nadu, became India's first transgender (transwoman) cricket umpire, made possible after the Tamil Nadu Cricket Association introduced an "Other" gender category in its State Panel Umpire examination application.
- **Pair 2 is correctly matched:** Squadron Leader Sanya became the first woman officer in the Indian Air Force's 93-year history to earn the coveted Category-A Qualified Flying Instructor (QFI) rating, the highest-level instructor qualification, responsible for training even other instructors.
- **Pair 3 is correctly matched:** Professor Kuljeet Kaur Marhas of the Physical Research Laboratory (PRL), Ahmedabad, became the first Indian woman elected Fellow of The Meteoritical Society (for 2026), and only the third Indian scientist overall to receive this honour, after Devendra Lal and J.N. Goswami. She is known for her work on short-lived radionuclides and pre-solar grains in meteorite samples, including from the Stardust, Hayabusa, and Apollo missions.
- **Pair 4 is correctly matched:** Christina Koch was the only woman among the four-member Artemis II crew (alongside Reid Wiseman, Victor Glover, and Jeremy Hansen), becoming the first woman to travel around the Moon during the mission's lunar flyby (1–10 April 2026).

Q 126. Ans. B

Explanation:

- **SEBI introduced the SWAGAT-FI framework, full form:** Single Window Automatic & Generalised Access for Trusted Foreign Investors, via two notifications, amending both the FPI (Foreign Portfolio Investors) Regulations, 2019, and the FVCI (Foreign Venture Capital Investors) Regulations, 2000. The framework, approved at SEBI's 211th Board meeting (12 September 2025), comes into force from 1 June 2026.
- **Key features:**
  - ✓ Built on a "trust-but-verify" approach for sophisticated, low-risk foreign investors — including sovereign wealth funds, central banks, government-backed funds, multilateral institutions, regulated public retail funds, pension funds, and appropriately regulated insurance companies.
  - ✓ **Enables dual registration:** eligible investors registered as FPIs can also register as FVCIs through a single, unified process without additional documentation — allowing investment in listed equities/debt (as FPI) as well as unlisted companies/startups in specified sectors (as FVCI).
  - ✓ Extends the registration/renewal and KYC review cycle from the earlier 3–5 years to 10 years.
  - ✓ SWAGAT-FI-classified investors already account for over 70% of total assets held by India's 11,913+ registered FPIs (as of June 2025), worth ₹80.83 lakh crore.

Q 127. Ans. D

Explanation:

- **Statement 1 is correct:** The Minuteman III intercontinental ballistic missile (ICBM) is a United States-developed weapon system, forming the land-based leg of the American nuclear triad, in service since 1970.
- **Statement 2 is correct:** Russia tested its nuclear-powered, nuclear-capable "Poseidon" underwater drone on 28 October 2025, as announced by President Vladimir Putin. It marked the first time the drone's nuclear reactor was activated during a test — coming just days after Russia also tested its nuclear-powered "Burevestnik" cruise missile. Putin described Poseidon as unmatched in speed and depth, and effectively "uninterceptable."
- **Statement 3 is correct:** South Korea unveiled the Hyunmoo-5 — nicknamed the "monster missile" — during its 76th Armed Forces Day ceremony on 1 October 2024, with deployment to frontline units beginning in late 2025. Developed by the Agency for Defense Development with Hanwha Aerospace, it carries an unprecedented 8-ton conventional warhead designed to destroy North Korea's deeply buried underground bunkers, and forms a key part of South Korea's "Korea Massive Punishment and Retaliation" (KMPR) deterrence doctrine.

Q 128. Ans. B

Explanation:

- Tirumala Hills (Tirupati district) and Erra Matti Dibbalu or Red Sand Dunes (near Visakhapatnam), both in Andhra Pradesh, were added to UNESCO's Tentative List of World Heritage Sites in September 2025, along with five other Indian natural sites (Deccan Traps at Panchgani-Mahabaleshwar, Geological Heritage of St. Mary's Island Cluster, Meghalayan Age Caves, Naga Hill Ophiolite, and Natural Heritage of Varkala). This took India's total tentative-list properties to 69 (49 Cultural, 3 Mixed, 17 Natural).
- **Additional Facts:**
  - ✓ **Erra Matti Dibbalu:** A ~1,500-acre coastal red-sand-dune formation, declared a National Geo-heritage Monument by the Geological Survey of India (GSI) in 2016, first documented by geologist William King in 1886. It qualifies under UNESCO's "Tectonic and Coastal Systems" theme.
  - ✓ **Tirumala Hills:** Known for the Eparchaean Unconformity — a geological boundary where 2.5-billion-year-old rocks meet younger Proterozoic formations — plus the ancient Silathoranam natural arch, and rich biodiversity within the Seshachalam Biosphere Reserve. It qualifies under UNESCO's "History of Planet Earth and Evolution of Life" theme.

Q 129. Ans. D

Explanation:

- **Mera Bharat Mera Yogdan:** Delhi (The campaign was launched by the Delhi Government to promote fuel conservation and public participation through measures like using public transport).
- **Jan Bhagidari Sabse Door Sabse Pehle:** Chhattisgarh (A campaign launched by the Chhattisgarh government to ensure the delivery of welfare schemes and essential services to remote tribal villages).
- **Mushroom Mission:** Andhra Pradesh (The AP government launched the Mushroom Mission with an estimated investment of ₹13,000 crore to boost agricultural income and become India's largest mushroom producer).
- **Ragi Java for school students:** Andhra Pradesh (Under state school nutrition initiatives, the government recently extended its agreement with the Sri Sathya Sai Central Trust to distribute nutritious Ragi Java to students).
- **Additional Facts:**
  - ✓ **Assam (Mukhya Mantri Mahila Udyamita Abhiyan):** Launched to promote entrepreneurship and self-employment among rural women by providing financial grants to members of self-help groups (SHGs).
  - ✓ **Kerala (Jyothi Scheme):** A targeted educational initiative designed specifically to enroll the children of migrant workers into the state's public schools and Anganwadi centers, ensuring they do not fall out of the formal education system.
  - ✓ **Andhra Pradesh (Zero Poverty – P4 Policy):** An ambitious long-term policy framework rolled out to completely eradicate poverty in the state by 2047 through a public-private partnership model (P4).
  - ✓ **Uttar Pradesh (UP AGREES):** The UP Agriculture Growth and Rural Enterprise Ecosystems project, backed by the World Bank, was launched to transform the state's agricultural sector and digitally empower rural youth.

- ✓ **Mizoram (Bana Kaih Scheme):** Also known as the "Handholding" scheme, this provides targeted financial, technical, and institutional support to local entrepreneurs and farmers to help them become economically self-sufficient.
- ✓ **Odisha (Shahid Madho Singh Haath Kharcha Scheme):** Aimed at reducing dropout rates among tribal students, it provides a one-time financial incentive of ₹5,000 to ST students enrolling in Classes IX and XI.

Q 130. Ans. C

Explanation:

- Reliance Chairman Mukesh Ambani unveiled Jio's "People First" AI platform at the Vibrant Gujarat Regional Conference 2026 in Rajkot on 11 January 2026, describing it as an indigenous artificial intelligence platform "built in India, for India and the world."
- **Key features:**
  - ✓ The "people-first" philosophy prioritises everyday users and usability over complex enterprise-only tooling, aiming for inclusive, affordable, local-language AI access across India.
  - ✓ It is backed by what Reliance calls India's largest AI-ready data centre, being built in Jamnagar, Gujarat, intended to serve as the computing backbone for scalable, high-speed, locally-hosted AI processing, reducing dependence on foreign cloud infrastructure.
  - ✓ This positions Jio's platform as distinct from its earlier enterprise-focused offering, JioBrain (unveiled 2024, for telecom network optimisation and enterprise AI-as-a-service), and its collaboration with IIT Bombay on BharatGPT, "People First" is framed as Jio's consumer-facing, nationally-built AI platform.

Q 131. Ans. B

Explanation:

- **Pair 1 is correctly matched:** Sanjay Jamuar, an ex-IRTS officer and DMRC's first-ever O&M employee (1998), was appointed the first CEO of Delhi Metro International Limited (DMIL) in April 2026 — a new DMRC subsidiary created to export India's metro expertise globally.
- **Pair 2 is correctly matched:** Dilip Kumar, a senior IRSME (Indian Railway Service of Mechanical Engineers) officer, assumed charge as Chief Vigilance Officer of SAIL on 27 April 2026, approved by the Appointments Committee of the Cabinet for a three-year tenure.
- **Pair 3 is incorrectly matched: Lt. General N.S. Raja Subramani (Retd.),** not Shashi Shekhar Vempati, became India's 3rd Chief of Defence Staff on 31 May 2026. Shashi Shekhar Vempati is instead known as the former CEO of Prasar Bharati (2017–2022). Shashi Shekhar Vempati was appointed as the Chairperson of the Central Board of Film Certification (CBFC) on May 6, 2026.
- **Pair 4 is incorrectly matched:** N.S. Raja Subramani holds the CDS post, not the Chairpersonship of the Central Board of Film Certification (CBFC), this pairing incorrectly swaps his actual designation.

Q 132. Ans. C

Explanation:

- **Gruha Lakshmi Portal 2 (Karnataka Government):** A state welfare scheme portal providing monthly financial assistance to women heads of eligible households in Karnataka.
- **Sujalam India App 1 (Ministry of Jal Shakti):** A central government app related to water conservation/management initiatives under the Ministry of Jal Shakti.
- **Gaja Rakshak App 3 (Madhya Pradesh Government):** A state government app aimed at human-elephant conflict management/wildlife protection in Madhya Pradesh.
- **Pratibha Setu Portal 4 (Union Public Service Commission, UPSC):** A UPSC initiative/portal connecting talented, meritorious candidates who narrowly missed selection with other recruiting organisations and opportunities.

Q 133. Ans. B

Explanation:

- **Pair 1 is incorrectly matched:** ASEM (Asia-Europe Meeting) has no permanent secretariat or headquarters — it is an informal dialogue process coordinated jointly by foreign ministries, not a formal institution. Its only permanently established body, the Asia-Europe Foundation (ASEF), is based in Singapore, not Brussels.
- **Pair 2 is correctly matched:** BIMSTEC's permanent secretariat is located in Dhaka, Bangladesh, established in 2014.
- **Pair 3 is correctly matched:** The SCO Secretariat is headquartered in Beijing, China.
- **Pair 4 is correctly matched:** The International Criminal Court is headquartered in The Hague, Netherlands.

Q 134. Ans. C

Explanation:

- The 24th IORA Council of Ministers Meeting was held virtually on 21 May 2025, hosted and chaired by Sri Lanka, under the theme "Sustainable Indian Ocean for Future Generations." India's delegation at this meeting was led by Secretary (East) P. Kumaran of the Ministry of External Affairs, rather than the Foreign Minister, Foreign Secretary, or NSA — reflecting the working/senior-official level of this particular Council of Ministers session.
- **Key points:**
  - ✓ The meeting focused on sustainable development, blue economy cooperation, and long-term environmental stewardship across IORA's Indian Ocean member states.
  - ✓ India reiterated its commitment to regional maritime cooperation, security, and collective economic development of the Indian Ocean Rim.
  - ✓ Sri Lanka's chairmanship was set to be followed by India taking over the IORA Chairship for the 2025–27 term.

Q 135. Ans. C

Explanation:

- The **National Youth Festival 2026** (officially reimagined as the Viksit Bharat Young Leaders Dialogue) was held at **Bharat Mandapam, New Delhi** from January 9 to 12, 2026.
- Prime Minister Narendra Modi addressed the valedictory session on January 12 (National Youth Day, commemorating Swami Vivekananda's birth anniversary).
- **Additional Facts:**
  - ✓ **Viksit Bharat Young Leaders Dialogue (VBYLD):** Starting in 2025, the traditional National Youth Festival format was restructured into the VBYLD to actively align with the vision of making India a developed nation by 2047 (Viksit Bharat @2047).
  - ✓ **MY Bharat Platform:** The selection process for the 2026 festival involved a national-level quiz conducted on the Mera Yuva Bharat (MY Bharat) portal. Over 50 lakh youths participated in the initial rounds, eventually culminating in a final shortlist of delegates.
  - ✓ **Key Digital Initiatives Showcased:** During the 2026 NYF, the Ministry of Education highlighted transformative digital platforms. This included the **Rashtriya e-Pustakalaya** (a rich digital repository of books), **e-Jaadui Pitara** (play-based digital content for foundational learning), and the **My Career Advisor app** for employability guidance.
  - ✓ **ULLAS & Vidyanjali:** The 2026 event heavily promoted the ULLAS (Understanding Lifelong Learning for All in Society) scheme for adult education and the Vidyanjali portal, which encourages professionals and alumni to volunteer in government schools.
  - ✓ **Previous Venues:** For historical context in MCQs, the National Youth Festival was held in **Nashik**, Maharashtra in 2024, and **Hubballi-Dharwad**, Karnataka in 2023.

Q 136. Ans. A

Explanation:

- A major breakthrough at the UN Climate Change Conference (COP30) in Belém, Brazil, was the agreement to at least triple adaptation finance for developing nations by 2035.
- The overarching climate finance goal, the New Collective Quantified Goal (NCQG), was established at COP29 in Baku at $1.3 trillion annually from all sources (with at least $300 billion in core public finance) by 2035. COP30 focused on operationalizing this $1.3 trillion pathway rather than upgrading the target.
- **Additional Facts:**
  - ✓ **The Global Mutirão:** The final overarching agreement at COP30 (November 2025) was officially termed the Mutirão Decision. "Mutirão" is a Brazilian Portuguese term signifying a collective mobilization or joint effort, underscoring the Belém summit's branding as the "COP of Implementation".
  - ✓ **Just Transition Mechanism:** A historic win for labor and civil society at COP30 was the formal adoption of a "Just Transition Mechanism". This framework aims to financially support and protect workers, frontline communities, and developing countries as they shift toward low-carbon economies.
  - ✓ **Tropical Forests Forever Facility (TFFF):** Championed heavily by the Brazilian presidency, this massive global fund to reward rainforest nations for maintaining standing forests received billions in initial pledges during the Belém summit.
  - ✓ **Voluntary Roadmaps:** Because fossil fuel phase-out

was blocked in the official UN consensus, Brazil's COP Presidency launched two voluntary, extra-UN roadmaps to keep the momentum alive: one for halting deforestation and another for transitioning away from fossil fuels.

- ✓ **COP31 Host:** Following diplomatic negotiations where Australia agreed to back its bid, **Antalya, Türkiye (Turkey)** was officially announced as the host city for the upcoming COP31 in November 2026.

Q 137. Ans. C

Explanation:

- **"The One: Cricket, My Life and More"** is the autobiography of former Indian opening batsman **Shikhar Dhawan**.
- Co-authored with Namita Kala and Chandresh Narayanan, and released in July 2025.
- **Additional Facts:**
  - ✓ **The Test of My Life:** Yuvraj Singh (chronicles his battle with cancer and return to the game)
  - ✓ **A Century is Not Enough:** Sourav Ganguly
  - ✓ **Playing It My Way:** Sachin Tendulkar
  - ✓ **281 and Beyond:** V.V.S. Laxman
  - ✓ **Believe:** Suresh Raina
  - ✓ **Stargazing: The Players in My Life:** Ravi Shastri
  - ✓ **No Spin:** Shane Warne

Q 138. Ans. C

Explanation:

- The opening match of the 2026 FIFA World Cup was hosted at the legendary Estadio Azteca (referred to as Mexico City Stadium during the tournament) in Mexico City on June 11, 2026.
- **Additional Facts:**
  - ✓ **Host Nations:** For the first time in history, the tournament is being jointly hosted by three countries: the **United States US, Canada CA, and Mexico MX**.
  - ✓ **Expanded Format:** This edition marks the first time the tournament has expanded to include **48 participating teams**, a significant jump from the traditional 32 teams.
  - ✓ **Matches & Venues:** Because of the expanded format, a record **104 matches** are being played across **16 different host cities**,.
  - ✓ **Group Stage & Knockouts:** The 48 teams were divided into 12 groups of four. The tournament introduced a new **Round of 32** knockout phase, accommodating the top two teams from each group plus the eight best third-place finishers.
  - ✓ **The Final Match:** The 2026 World Cup Final is scheduled for **July 19, 2026**, at **MetLife Stadium** (New York/New Jersey Stadium) in the United States.

Q 139. Ans. A

Explanation:

- India's Esha Singh delivered a historic performance at the ISSF World Cup 2026 in Munich, winning the **gold medal** in the women's 25m pistol event.
- She secured the top spot on the podium with a stunning score of 43 hits out of 50 in the final.
- Germany's Doreen Vennekamp settled for the silver medal with 38 hits, and Bulgaria's Miroslava Mincheva took the bronze.
- **Additional Facts:**
  - ✓ **World Cup Final Qualification:** By winning this gold, the 21-year-old shooter earned a direct qualification spot for the season-ending ISSF World Cup Final, scheduled to be held in Rome, Italy, in December 2026.
  - ✓ **Double Medal Joy:** Esha Singh also secured a **silver medal** in the women's 10m air pistol event at the same Munich World Cup. In that event, her compatriot **Suruchi Singh** clinched the gold, completing a brilliant 1-2 finish for India.
  - ✓ **Significance:** Esha's 25m pistol victory marked **India's first individual gold medal** of the ISSF World Cup season across all rifle, pistol, and shotgun events.
  - ✓ **High-Profile Defeats:** Her record-breaking 25m final was particularly impressive because it featured reigning Olympic champion Yang Ji-in, who had topped the qualification rounds with a score of 592-28x but ultimately struggled in the medal round and finished fifth.

Q 140. Ans. B

Explanation:

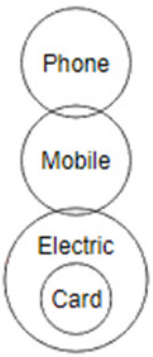
- The **'New Delhi Declaration'** is the first-ever global declaration exclusively focused on the conservation of all seven major big cat species.
- It is the primary outcome of the inaugural International Big Cat Alliance (IBCA) Summit, which was hosted by India in New Delhi on June 1–2, 2026.
- The declaration is designed to move big cat conservation beyond isolated protected areas by creating a common global framework for

landscape-based habitat protection, maintaining safe wildlife corridors, and strengthening transboundary cooperation against poaching and illegal wildlife trade.

- **Additional Facts:**
  - ✓ **The 7 Big Cats:** The IBCA focuses on protecting seven apex predators: Tiger, Lion, Leopard, Snow Leopard, Cheetah, Jaguar, and Puma. Remarkably, India is the only country in the world that is naturally home to five of these seven species (excluding the Jaguar and Puma).
  - ✓ **IBCA Origins:** The International Big Cat Alliance was originally launched by Prime Minister Narendra Modi in 2023 to commemorate the 50th anniversary of Project Tiger. Following Cabinet approval, its permanent headquarters and Secretariat have been established in New Delhi.
  - ✓ **Summit Theme:** The guiding theme for the 2026 IBCA Summit was "Save Big Cats, Save Humanity, Save Ecosystem".
  - ✓ **Constitutional Mandate:** In the context of Indian polity exams, such conservation diplomacy is strongly backed by **Article 51A(g)** of the Constitution, which casts a Fundamental Duty on every citizen to protect and improve the natural environment and wildlife.

Q 141. 141. Ans. A

Explanation:



- **Checking C1:** All Cards can be Phone
- We can see that in the Venn diagram, there is no negative statement is given against the possibility. Therefore, C1 follows.
- **Checking C2:** Some Electric are Phone
- We can see that in the Venn diagram, no direct relation is given between the two. Therefore, C2 does not follow.
- Hence, option A is correct.

Q 142. Ans. B

Explanation:

- **Using the above information, we get the following arrangement:**

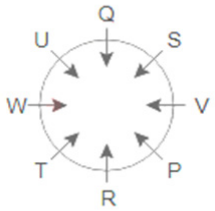
8	N
7	O
6	R
5	S
4	T
3	Q
2	P
1	M

- We get "Two people live between Q and R.
- Hence, Option B is correct.

Q 143. Ans. C

Explanation:

Using the above information, we get the following arrangement:



We get "P" sitting second to the left of S.
Hence, Option C is correct.

Q 144. Ans. C

Explanation:

Given that John is 26 ranks ahead of Kamal and Kamal's rank is 17th from the last.
So, John's rank = (26 + 17) = 43rd from the last.
Therefore, the rank of John from the top is = 64 – 43 + 1 = 22
Hence, option C is correct.

Q 145. Ans. A

Explanation:

- Statement I (government increasing tax rates on luxury goods) is the cause, and Statement II (the decrease in sales of luxury cars) is its effect.
- The increase in tax rates makes luxury cars more expensive for consumers, leading to a decrease in demand and, consequently, a decline in sales.
- Hence, option A is correct.

Q 146. Ans. C

Explanation:

- According to the question,
- Time taken to cover a distance of 108 km = 90 minutes = 3/2 hours
- Speed of the bike = 108 ÷ (3/2) = 108 × 2/3 = 72 km/hr
- New speed = 72 - 24 = 48 km/hr
- Required time = 108/48 = 9/4 hours
- = 9/4 × 60 minutes
- = 135 minutes

Q 147. Ans. D

Explanation:

- According to the question,
- Printer = P and Scanner = S
- **First condition:**
- ⇒ 3P + S = ₹82,000 .....(1)
- **Second condition:**
- ⇒ P + 2S = ₹74,000 .....(2)
- **Multiplying equation (2) by 3:**
- ⇒ 3P + 6S = ₹222,000 .....(3)
- **Subtracting equation (1) from equation (3):**
- ⇒ (3P + 6S) - (3P + S) = ₹222,000 - ₹82,000
- ⇒ 5S = ₹140,000
- ⇒ S = ₹28,000
- Therefore, the cost of one scanner is ₹28,000.

Q 148. Ans. A

Explanation:

- Given,
- ⇒ **(7x + 4y) :** (7x - 4y) = 13 : 5
- By the componendo and dividendo rule,
- $\Rightarrow \frac{(7x+4y)+(7x-4y)}{(7x+4y)-(7x-4y)} = \frac{13+5}{13-5} =$
- $\Rightarrow \frac{14x}{8y} = \frac{18}{8}$
- $\Rightarrow \frac{7x}{4y} = \frac{9}{4}$
- ⇒ **x :** y = 9 : 7

Q 149. Ans. A

Explanation:

- √1024 - 6/5 of 372.5 + 20% of 760
- 32 - 65 × 372.5 + 15 × 760
- 32 - 447 + 152
- -263

Q 150. Ans. C

Explanation:

- According to the question,
- Quotient = 18
- Divisor = 20 × 18 = 360
- Divisor = 4 × Remainder
- ⇒ Remainder = 360/4 = 90
- Now,
- Dividend = (Divisor × Quotient) + Remainder
- ⇒ Dividend = (360 × 18) + 90
- ⇒ 6480 + 90
- ⇒ 6570
- Therefore, Dividend = 6570

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